



ADMS

16.25.1

OMS Interface Server Programmer Guide

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1. Introduction

The OMS Interface server provides a standard web server interface to allow third-party applications to communicate with the OMS server (NetServer).

The OMS Adapter provides a web services portal for messages among the OMS Interface server and external applications, including the Outage Assist solution.

The Customer Experience Adapter, Automated Notification Paging Services Adapter, and View Outage Portal Adapter use the same approach as the OMS Adapter to send data to external applications.

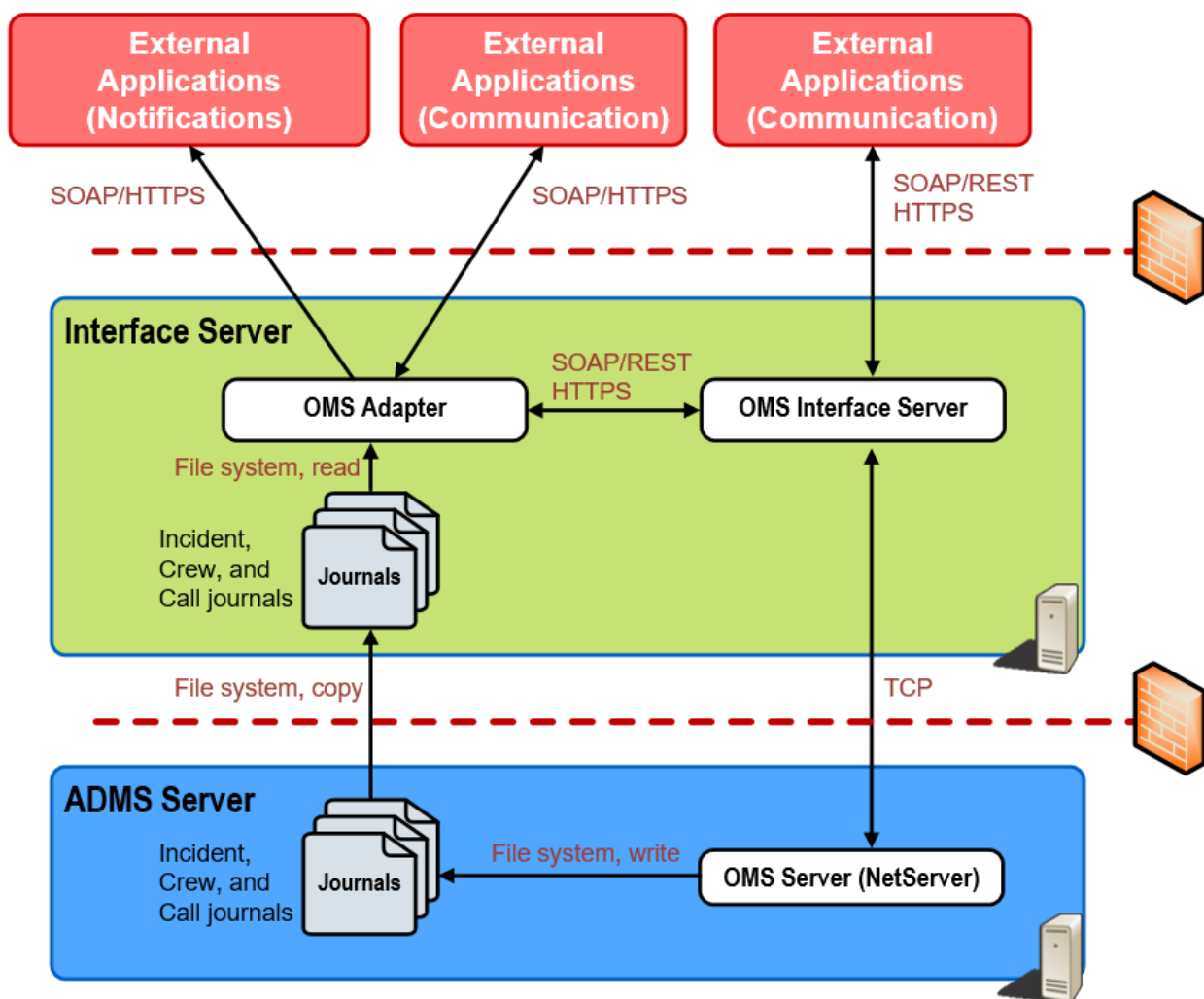


Figure 1. OMS Interface Server and OMS Adapter

An example of the OMS Interface server and OMS Adapter flow is the following:

1. An incident appears in ADMS.
2. NetServer creates an incident record in the Incident Journal.

3. NetServer sends incident data to the OMS Interface server.
4. OMS Interface server receives the GetIncidentByID event at port 8802 (WebServiceBaseURL).
5. OMS Adapter receives an incident creation event based on the Incident Journal record.
6. OMS Adapter queries the OMS Interface server at port 8802 (OmsServiceURL) for the incident data and receives the data.
7. OMS Adapter sends a CreatedIncidentEvent notification to Outage Assist at port 9090 (OmsEsAdapterURL).
8. Customer Experience Adapter sends an IncidentUpdateNotification to the Customer Experience web service client at `http://localhost:8447/api/ext/ce/v2/IncidentUpdate (BaseUri + IncidentUpdateUri)`.

1.1. Process Management

The OMS Interface server and OMS Adapter do not have a CFGRPROXY interface. They can be managed by CFGMAN and PROCMAN to provide redundancy and high availability. The detailed configuration depends on the deployment needs. For more information, refer to the *Habitat Configuration Manager User Guide* and the *PROCMAN User's Guide*.

1.2. Security

The OMS Interface server and OMS Adapter can be run using standard HTTP or HTTPS. To provide additional security for web service messages, an "authorization" header can be used for requests. In addition, OMS Adapter can include trusted clients with encrypted passwords.

The OMS Interface server also supports the UAA authentication and using SSL certificates for the HTTPS mode.

Authorization Header

To provide additional security for control service calls, an "authorization" header is used to authenticate calls from external applications for all POST update requests.

The authorization header format is as follows:

```
"Authorization: CustomAuthHeader oQWFe7pQj/UUTiBpRwt63Efo1aQ="
```

The authorization header includes the hashed key, which is calculated by the request URI, random value (salt), and password in the OMS Adapter configuration file.

The following cryptographic hash algorithms are supported: MD5, SHA1, SHA256, SHA384, and SHA512. If another value is specified, MD5 is used by default.

1.3. Data Flow Restrictions

Using web services means that ADMS and external systems do not maintain a constant connection. Each request is a new connection. To restrict connections originating from less secure zones, all message exchanges between ADMS and an external system is initiated by ADMS.

2. OMS Interface Server

The OMS Interface server provides a standard web server interface to allow third-party applications to communicate with the OMS server (NetServer). The OMS Interface server acts as a client to the OMS server, using the same protocol that the Browser client uses to communicate with the OMS server. This enables access to the same set of OMS data (for example, incident and crew data) as the client. OMS Interface server can also connect to the DMS Interface server to change device data and perform DMS queries.

The OMS Interface server does not use NetDynCtrl to connect to the OMS server; therefore, it does not need dynamics information.

Multiple instances of the OMS Interface server can run simultaneously.

The OMS Interface server routes OMS data from NetServer to the OMS Adapter and external applications, and also routes OMS data from the OMS Adapter and external applications to NetServer.

The OMS Services web service allows external applications to communicate with the OMS server (NetServer) to query OMS and update switch statuses. The OMS Services web service is implemented using REST.

The OMS Interface server supports a plugin web service, which allows querying incident data from the NetServer.

2.1. REST Web Service

Important

The REST web service is used internally. The information is provided here for reference purposes only.

The OMS Services REST web service is provided through the following home URI, which lists all available service calls, entity types, bindings, etc.:

`http://<YourServerIP>:<Port>/oms/services/`

For example:

`http://localhost:8802/oms/services/`

For reference, the REST service messages are listed in the table below. For details about REST messages, see <http://localhost:8802/oms/services/rest/help>.

The request and response format for the REST service messages is `WebMessageFormat.Json`.

Table 1. REST Service Messages

Name	Protocol	Method	Type	Description
CallPriorityConfig	REST	GET	Input	Gets the call priority configuration from the OMS server.
CanClearIncident	REST	GET	Input	Checks whether an incident can be cleared according to the validation rules.
CreatePlannedOutage	REST	POST	Input	Creates planned outages. • If there are any delimiting devices, they are displayed with distinct icons. • Delimiting devices and their associated customers are not considered part of the outage, as they are backfed through external devices such as jumpers.
DeleteAreaNote	REST	POST	Input	Deletes an area setting.
DeleteCall	REST	GET	Input	Deletes a customer call or an unassociated call.
GetAllAreaNotes	REST	GET	Input	Queries the OMS server for data about all area settings.
GetAllCrews	REST	GET	Input	Queries the OMS server for data about all crews.
GetAreaNotesByIds	REST	POST	Input	Queries the OMS server for data about area settings by area setting IDs.
GetCrewsByIds	REST	POST	Input	Queries the OMS server for data about crews by crew IDs.
GetCustomersForDevice	REST	GET	Input	Gets a list of customers that are downstream of a device.
GetCustomFieldConfiguration	REST	GET	Input	Gets filtered custom field configurations for an object type (incident, order, damage assessment, crew, area setting, or planned outage).
GetCustomFieldConfigurationAll	REST	GET	Input	Get all custom field configurations for an object type (incident, order, damage assessment, crew, area setting, or planned outage).
/GetDamages	REST	GET	Input	Gets a list of damage objects.
/GetDamage?id={id}	REST	GET	Input	Gets a damage object by ID.
GetDownStreamIncidentList	REST	GET	Input	Gets a list of downstream incidents for an incident.
GetDownStreamIncidents	REST	GET	Input	Gets downstream incidents.
GetDownStreamOrderList	REST	GET	Input	Gets downstream orders by incident ID.
GetDownStreamOrderListFromDevice	REST	GET	Input	Gets downstream orders by device ID.
GetIncidentByName	REST	GET	Input	Gets an incident by name.
GetIncidentsByIds	REST	POST	Input	Gets incident by IDs.
GetOrdersOnFeeder	REST	GET	Input	Requests the list of orders on the feeder that is associated with the specified device.
GetStructureById	REST	GET	Input	Returns structure data by ID.
GetWorkOrdersForIncident	REST	GET	Input	Returns active orders for incident by ID.

Name	Protocol	Method	Type	Description
Health	REST	GET	Input	Checks whether the service is up and running.
Incidents	REST	GET	Input	Queries the OMS server for all incident data.
UpdateRestorationStep	REST	POST	Input	Sends a request to the OMS server to update a restoration step for an active incident.
UpdateSwitchStatus	REST	POST	Input	Sends a request to the OMS server to update a switch state.

Rest URLs:

- <http://localhost:8802/oms/services/rest/CallPriorityConfig>
- <http://localhost:8802/oms/services/rest/CanClearIncident?Id={ID}&user={USERNAME}>
- <http://localhost:8802/oms/services/rest/CreatePlannedOutage>
- <http://localhost:8802/oms/services/rest/DeleteAreaNote>
- http://localhost:8802/oms/services/rest/DeleteCall?user={USERNAME}&call_mRID={MRID}
- <http://localhost:8802/oms/services/rest/GetAllAreaNotes>
- <http://localhost:8802/oms/services/rest/GetAllCrews>
- <http://localhost:8802/oms/services/rest/GetAreaNotesByIds>
- <http://localhost:8802/oms/services/rest/GetCrewsByIds>
- <http://localhost:8802/oms/services/rest/GetCustomersForDevice?station={STATION}&device={DEVICE}>
- <http://localhost:8802/oms/services/rest/GetCustomFieldConfiguration?ObjectType={OBJECTTYPE}&AppType={APPTYPE}&IncludeEmptyAppType={INCLUDEEMPTYAPPTYPE}>

Where, ObjectType can be Incident, Order, DamageAssessment, Crew, AreaSettings, or PlannedOutage; AppType (optional), can be an Application Type value specified in the OMS Server Configuration Editor.

- <http://localhost:8802/oms/services/rest/GetCustomFieldConfigurationAll?ObjectType={OBJECTTYPE}>
- <http://localhost:8802/oms/services/rest/GetDamages>
- <http://localhost:8802/oms/services/rest/GetDamage?id={id}>
- <http://localhost:8802/oms/services/rest/GetDownStreamIncidentList?master={MASTERID}&user={USRNAME}>
- <http://localhost:8802/oms/services/rest/GetDownstreamIncidents?mRID={MRID}&user={USERNAME}&time={DATETIME}>
- <http://localhost:8802/oms/services/rest/GetDownStreamOrderList?masterIncident={MASTERINCIDENT}&masterOrder={MASTERORDER}&user={USRNAME}>
- <http://localhost:8802/oms/services/rest/GetDownStreamOrderListFromDevice?deviceId={deviceId}&deviceType={deviceType}&station={station}&feederName={feederName}&user={userName}>

- `http://localhost:8802/oms/services/rest/GetIncidentByName?incidentName={incidentName}`
- `http://localhost:8802/oms/services/rest/GetIncidentsByIds`
- `http://localhost:8802/oms/services/rest/GetOrdersOnFeeder?stationName={stationName}&deviceId={deviceId}&user={usrName}`
- `http://localhost:8802/oms/services/rest/GetStructureById?id={ID}`
- `http://localhost:8802/oms/services/rest/GetWorkOrdersForIncident?incidentId={INCIDENTID}`
- `http://localhost:8802/oms/services/rest/Health`
- `http://localhost:8802/Oms/services/rest/Incidents`
- `http://localhost:8802/oms/services/rest/UpdateRestorationStep`
- `http://localhost:8802/oms/services/rest/UpdateSwitchStatus`

2.1.1. Request Schemas

This section provides schemas for OMS Interface server REST requests.

2.1.1.1. DeleteAreaNote

The request body is a string with an area setting ID. For example:

```
"RAVEN"
```

2.1.1.2. GetAreaNotesByIds

The request must contain the list of area setting IDs. For example:

```
[
  "PEBBLE BEACH",
  "RAVEN"
]
```

The request schema:

```
<xs:schema xmlns:tns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
  elementFormDefault="qualified"
  targetNamespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfstring">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="string" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfstring" nillable="true" type="tns:ArrayOfstring" />
</xs:schema>
```

```
</xs:schema>
```

2.1.1.3. GetCrewsByIds

```
<xs:schema
xmlns:tns="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
elementFormDefault="qualified"
targetNamespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="StringListRequest">
    <xs:sequence>
      <xs:element name="NetworkIncidentManagementServiceContext" nillable="true"
type="tns:RequstContext" />
      <xs:element name="StringList" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="StringListRequest" nillable="true" type="tns:StringListRequest" />
  <xs:complexType name="RequstContext">
    <xs:sequence>
      <xs:element minOccurs="0" name="correlationID" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="clientID" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="callbackHttpURL" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="userName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="timeSent" type="xs:dateTime" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="RequstContext" nillable="true" type="tns:RequstContext" />
</xs:schema>
```

2.1.1.4. UpdateRestorationStep

```
<xs:schema
xmlns:tns="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
elementFormDefault="qualified"
targetNamespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="UpdateRestorationStepRequest">
    <xs:sequence>
      <xs:element name="Fields" nillable="true" type="tns:ArrayOfFieldUpdateData" />
      <xs:element name="RestorationStepId" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="UpdateRestorationStepRequest" nillable="true"
type="tns:UpdateRestorationStepRequest" />
  <xs:complexType name="ArrayOfFieldUpdateData">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="FieldUpdateData" nillable="true"
type="tns:FieldUpdateData" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="FieldUpdateData" nillable="true"
type="tns:FieldUpdateData" />
</xs:schema>
```

```

type="tns:FieldUpdateData" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfFieldUpdateData" nillable="true" type="tns:ArrayOfFieldUpdateData" />
<xs:complexType name="FieldUpdateData">
  <xs:sequence>
    <xs:element name="FieldName" nillable="true" type="xs:string" />
    <xs:element name="NewValue" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OldValue" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="FieldUpdateData" nillable="true" type="tns:FieldUpdateData" />
</xs:schema>

```

2.1.1.5. UpdateSwitchStatus

```

<xs:schema
xmlns:tns="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
elementFormDefault="qualified"
targetNamespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="UpdateSwitchStatusContext">
    <xs:sequence>
      <xs:element minOccurs="0" name="WorkManagementServiceContext" nillable="true"
type="tns:WorkManagementServiceContext" />
      <xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DevicePosition" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StationName" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="UpdateSwitchStatusContext" nillable="true" type="tns:UpdateSwitchStatusContext" />
  <xs:complexType name="WorkManagementServiceContext">
    <xs:sequence>
      <xs:element minOccurs="0" name="correlationID" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="clientID" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="callbackHttpURL" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="userName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="timeSent" type="xs:dateTime" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="WorkManagementServiceContext" nillable="true"
type="tns:WorkManagementServiceContext" />
</xs:schema>

```

2.1.2. Response Schemas

This section provides schemas for OMS Interface server REST request responses.

2.1.2.1. CallPriorityConfig

```
<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfCallPriorityConfigEntry">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CallPriorityConfigEntry" nillable="true"
type="tns:CallPriorityConfigEntry" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfCallPriorityConfigEntry" nillable="true"
type="tns:ArrayOfCallPriorityConfigEntry" />
  <xs:complexType name="CallPriorityConfigEntry">
    <xs:sequence>
      <xs:element minOccurs="0" name="Priority" type="xs:int" />
      <xs:element minOccurs="0" name="CallType" type="xs:int" />
      <xs:element minOccurs="0" name="Description" type="xs:int" />
      <xs:element minOccurs="0" name="Condition1" type="xs:int" />
      <xs:element minOccurs="0" name="Condition2" type="xs:int" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="CallPriorityConfigEntry" nillable="true" type="tns:CallPriorityConfigEntry" />
</xs:schema>
```

2.1.2.2. CanClearIncident

```
<xs:schema
xmlns:tns="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
elementFormDefault="qualified"
targetNamespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
  <xs:complexType name="RequestCanClearResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="out" nillable="true" type="tns:CanClearOut" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="RequestCanClearResponse" nillable="true" type="tns:RequestCanClearResponse" />
  <xs:complexType name="CanClearOut">
    <xs:sequence>
      <xs:element minOccurs="0" name="ValidationMessages" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
      <xs:element minOccurs="0" name="ValidationStatus"
xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil" type="q2:ValidationResult" />
      <xs:element minOccurs="0" name="Errors" nillable="true" type="tns:ArrayOfResponseError" />
      <xs:element minOccurs="0" name="Acknowledgement" nillable="true" type="tns:ResponseAct" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="CanClearOut" nillable="true" type="tns:CanClearOut" />
  <xs:complexType name="ArrayOfResponseError">
```

```

<xs:sequence>
  <xs:element minOccurs="0" maxOccurs="unbounded" name="ResponseError" nillable="true"
type="tns:ResponseError" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfResponseError" nillable="true" type="tns:ArrayOfResponseError" />
<xs:complexType name="ResponseError">
  <xs:sequence>
    <xs:element minOccurs="0" name="errorMessage" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="creationTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" name="systemID" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ErrorObject" nillable="true" type="tns:ErrorObject" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ResponseError" nillable="true" type="tns:ResponseError" />
<xs:complexType name="ErrorObject">
  <xs:sequence>
    <xs:element minOccurs="0" name="InstanceIDs" nillable="true" type="tns:InstanceIDs" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ErrorObject" nillable="true" type="tns:ErrorObject" />
<xs:complexType name="InstanceIDs">
  <xs:sequence>
    <xs:element minOccurs="0" name="identifierName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="identifierValue" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="InstanceIDs" nillable="true" type="tns:InstanceIDs" />
<xs:complexType name="ResponseAct">
  <xs:sequence>
    <xs:element minOccurs="0" name="outcome" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="creationTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" name="description" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ResponseAct" nillable="true" type="tns:ResponseAct" />
</xs:schema>

```

2.1.2.3. CreatePlannedOutage

```

{
  "PlannedOutageRecord":{
    "AlternateEndTimeBinary":"\\/Date(1727883556217-0700)\\/",
    "AlternateStartTimeBinary":"\\/Date(1727879925016-0700)\\/",
    "CausesIncident":false,
    "CustomerAccountsNotify":[{
      "CustomerKey":"555337702555337702555337702",
      "Notify":true
    }],
    "DeviceId":"22234044",
    "DownstreamDevices":[{"DeviceId":"21588861"},{"DeviceId":"21608617"}],
  }
}

```

```

    "DeviceType":4,
    "EventType":"replacing poles",
    "OutageMessage":"Hello, this is Outage Message",
    "PlannedOutageType":0,
    "PrimaryEndTimeBinary":"\\/Date(1727912408283-0700)\\/",
    "PrimaryStartTimeBinary":"\\/Date(1727908789156-0700)\\/",
    "RepresentativeAddress":"Representative Address SE",
    "RepresentativeEmail":"Representative@Address.com",
    "RepresentativeName":"Representative Name",
    "RepresentativeNumber":"(425) 555-55-88",
    "Station":"PEBBLE BEACH",
    "UseAlternateDate":false,
    "CustomFields":[{"
      "PropertyName":"P01",
      "Value":"true"},
    {
      "PropertyName":"P02",
      "Value":"1"
    }, {
      "PropertyName":"P03",
      "Value":"TEST CUSTOM FIELDS - from OIS"
    }, {
      "PropertyName":"P04",
      "Value":"123000"
    }, {
      "PropertyName":"P05",
      "Value":"05/12/2021 12:30 PM"
    }, {
      "PropertyName":"P06",
      "Value":"Additional Custom Fields - not configured"
    }
  ],
  "authorName":"dmsall"
}

```

2.1.2.4. DeleteAreaNote

```

<xs:schema
  xmlns:tns="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
  elementFormDefault="qualified"
  targetNamespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="RequestResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="out" nillable="true" type="tns:ResponseOut" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="RequestResponse" nillable="true" type="tns:RequestResponse" />
  <xs:complexType name="ResponseOut">
    <xs:sequence>
      <xs:element minOccurs="0" name="Errors" nillable="true" type="tns:ArrayOfResponseError" />
    </xs:sequence>
  </xs:complexType>

```

```

        <xs:element minOccurs="0" name="forwardResponseIDs" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="Acknowledgement" nillable="true" type="tns:ResponseAct" />
    </xs:sequence>
</xs:complexType>
<xs:element name="ResponseOut" nillable="true" type="tns:ResponseOut" />
<xs:complexType name="ArrayOfResponseError">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="ResponseError" nillable="true"
type="tns:ResponseError" />
    </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfResponseError" nillable="true" type="tns:ArrayOfResponseError" />
<xs:complexType name="ResponseError">
    <xs:sequence>
        <xs:element minOccurs="0" name="errorMessage" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="creationTime" nillable="true" type="xs:dateTime" />
        <xs:element minOccurs="0" name="systemID" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="ErrorObject" nillable="true" type="tns:ErrorObject" />
    </xs:sequence>
</xs:complexType>
<xs:element name="ResponseError" nillable="true" type="tns:ResponseError" />
<xs:complexType name="ErrorObject">
    <xs:sequence>
        <xs:element minOccurs="0" name="InstanceIDs" nillable="true" type="tns:InstanceIDs" />
    </xs:sequence>
</xs:complexType>
<xs:element name="ErrorObject" nillable="true" type="tns:ErrorObject" />
<xs:complexType name="InstanceIDs">
    <xs:sequence>
        <xs:element minOccurs="0" name="identifierName" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="identifierValue" nillable="true" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:element name="InstanceIDs" nillable="true" type="tns:InstanceIDs" />
<xs:complexType name="ResponseAct">
    <xs:sequence>
        <xs:element minOccurs="0" name="outcome" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="creationTime" nillable="true" type="xs:dateTime" />
        <xs:element minOccurs="0" name="description" nillable="true" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:element name="ResponseAct" nillable="true" type="tns:ResponseAct" />
</xs:schema>

```

2.1.2.5. DeleteCall

The response schema is RequestResponse. For more details, see [DeleteAreaNote](#).

2.1.2.6. GetAllAreaNotes

```
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfAreaNote">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="AreaNote" nillable="true"
type="tns:AreaNote" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfAreaNote" nillable="true" type="tns:ArrayOfAreaNote" />
  <xs:complexType name="AreaNote">
    <xs:sequence>
      <xs:element minOccurs="0" name="AreaNoteCode" type="xs:int" />
      <xs:element minOccurs="0" name="AreaNoteId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AreaNoteStartTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="AreaNoteETR" type="xs:dateTime" />
      <xs:element minOccurs="0" name="StormName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="LevelName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="SystemMode" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ITRDisable" type="xs:boolean" />
      <xs:element minOccurs="0" name="CreationTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="OverrideCaseNote" type="xs:boolean" />
      <xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
      <xs:element minOccurs="0" name="SuppressReasonCode" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ParentID" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="UseParent" type="xs:boolean" />
      <xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Remarks" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="AreaNote" nillable="true" type="tns:AreaNote" />
</xs:schema>
```

2.1.2.7. GetAllCrews

```
<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import
namespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice" />
  <xs:complexType name="ArrayOfCrewData">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CrewData" nillable="true"
type="tns:CrewData" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfCrewData" nillable="true" type="tns:ArrayOfCrewData" />
  <xs:complexType name="CrewData">
    <xs:sequence>
      <xs:element name="Id" nillable="true" type="xs:string" />
      <xs:element name="Name" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
```

```

    <xs:element name="BaseLocation" nillable="true" type="tns:Location" />
    <xs:element name="ContactInfo" nillable="true" type="xs:string" />
    <xs:element name="CrewMembers" nillable="true" type="tns:ArrayOfCrewMember" />
    <xs:element name="CrewWorkStatus" nillable="true"
xmlns:q1="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
type="q1:ArrayOfCrewWorkStatus" />
    <xs:element name="CurrentLocation" nillable="true" type="tns:Location" />
    <xs:element name="Vehicles" nillable="true" type="tns:ArrayOfVehicle" />
    <xs:element name="IsMobileDispatchable" type="xs:boolean" />
    <xs:element name="IsTemporary" type="xs:boolean" />
    <xs:element name="Status" nillable="true" type="tns:Status" />
    <xs:element name="Type" nillable="true" type="xs:string" />
    <xs:element name="Radio" nillable="true" type="xs:string" />
    <xs:element name="VehicleId" nillable="true" type="xs:string" />
    <xs:element name="OperatingArea" nillable="true" type="xs:string" />
    <xs:element name="PrimaryEmployeeName" nillable="true" type="xs:string" />
    <xs:element name="PrimaryFirstName" nillable="true" type="xs:string" />
    <xs:element name="PrimaryLastName" nillable="true" type="xs:string" />
    <xs:element name="EmployeeIds" nillable="true" type="xs:string" />
    <xs:element name="OpCenter" nillable="true" type="xs:string" />
    <xs:element name="IncidentName" nillable="true" type="xs:string" />
    <xs:element name="OperatingAreaId" nillable="true" type="xs:string" />
    <xs:element name="TempAera" nillable="true" type="xs:string" />
    <xs:element name="Active" type="xs:boolean" />
    <xs:element name="Company" nillable="true" type="xs:string" />
    <xs:element name="Dept" nillable="true" type="xs:string" />
    <xs:element name="IsForeign" type="xs:boolean" />
    <xs:element name="CrewDivision" nillable="true" type="xs:string" />
    <xs:element name="DefaultCrewArea" nillable="true" type="xs:string" />
    <xs:element name="AutomaticDispatchId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IsContractor" type="xs:boolean" />
    <xs:element minOccurs="0" name="IsEvaluator" type="xs:boolean" />
    <xs:element minOccurs="0" name="ActiveEmployeesCount" type="xs:int" />
    <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
    <xs:element minOccurs="0" name="JobStatus" type="xs:int" />
    <xs:element minOccurs="0" name="ShiftStatus" type="xs:int" />
    <xs:element minOccurs="0" name="MobileCommunicationStatus" type="xs:int" />
    <xs:element minOccurs="0" name="AvailabilityStatus" type="xs:int" />
  </xs:sequence>
</xs:complexType>
<xs:element name="CrewData" nillable="true" type="tns:CrewData" />
<xs:complexType name="Location">
  <xs:sequence>
    <xs:element minOccurs="0" name="posX" type="xs:float" />
    <xs:element minOccurs="0" name="posY" type="xs:float" />
  </xs:sequence>
</xs:complexType>
<xs:element name="Location" nillable="true" type="tns:Location" />
<xs:complexType name="ArrayOfCrewMember">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CrewMember" nillable="true"
type="tns:CrewMember" />
  </xs:sequence>

```

```

</xs:complexType>
<xs:element name="ArrayOfCrewMember" nillable="true" type="tns:ArrayOfCrewMember" />
<xs:complexType name="CrewMember">
  <xs:sequence>
    <xs:element minOccurs="0" name="IsLeader" type="xs:boolean" />
    <xs:element minOccurs="0" name="Employee" nillable="true" type="tns:EmployeeData" />
  </xs:sequence>
</xs:complexType>
<xs:element name="CrewMember" nillable="true" type="tns:CrewMember" />
<xs:complexType name="EmployeeData">
  <xs:sequence>
    <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="FIRSTNAME" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="LASTNAME" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CREWID" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="RADIONUMBER" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ISEVALUATOR" type="xs:boolean" />
    <xs:element minOccurs="0" name="ISPRIMARY" type="xs:boolean" />
    <xs:element minOccurs="0" name="MDTFLAG" type="xs:boolean" />
    <xs:element minOccurs="0" name="PHONENUMBER" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="EmployeeTypeName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DefaultLeadName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewArea" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewDivision" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="DefaultCrewArea" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="AutomaticDispatchId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IsContractor" type="xs:boolean" />
    <xs:element minOccurs="0" name="IsForeign" type="xs:boolean" />
    <xs:element minOccurs="0" name="Company" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="EmployeeData" nillable="true" type="tns:EmployeeData" />
<xs:complexType name="ArrayOfVehicle">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="Vehicle" nillable="true"
type="tns:Vehicle" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfVehicle" nillable="true" type="tns:ArrayOfVehicle" />
<xs:complexType name="Vehicle">
  <xs:sequence>
    <xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Type" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="HasMobileDataTerminal" type="xs:boolean" />
    <xs:element minOccurs="0" name="HasGPS" type="xs:boolean" />
    <xs:element minOccurs="0" name="EmployeeId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ReportedTime" type="xs:dateTime" />
  </xs:sequence>
</xs:complexType>
<xs:element name="Vehicle" nillable="true" type="tns:Vehicle" />
<xs:complexType name="Status">
  <xs:sequence>
    <xs:element minOccurs="0" name="Value" type="xs:int" />
  </xs:sequence>
</xs:complexType>

```

```

    <xs:element minOccurs="0" name="dateTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="Remark" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Reason" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="Status" nillable="true" type="tns:Status" />
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="tns:PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfPropertyValue" nillable="true" type="tns:ArrayOfPropertyValue" />
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element name="PropertyName" nillable="true" type="xs:string" />
    <xs:element name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="PropertyValue" nillable="true" type="tns:PropertyValue" />
</xs:schema>

```

2.1.2.8. GetAreaNotesByIds

The request schema is ArrayOfAreaNote. For more details, see [GetAllAreaNotes](#).

2.1.2.9. GetCrewsByIds

The request schema is ArrayOfCrewData. For more details, see [GetAllCrews](#).

2.1.2.10. GetCustomersForDevice

```

<xs:schema xmlns:tns="http://schemas.datacontract.org/2004/07/NetUtil" elementFormDefault="qualified"
targetNamespace="http://schemas.datacontract.org/2004/07/NetUtil"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/" />
  <xs:complexType name="GetDeviceCustomersResponse">
    <xs:sequence>
      <xs:element name="Customers" nillable="true" type="tns:ArrayOfTracedCustomer" />
      <xs:element name="Status" nillable="true" type="xs:string" />
      <xs:element name="StatusCode" type="tns:NetResponseErrorEnum" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="GetDeviceCustomersResponse" nillable="true" type="tns:GetDeviceCustomersResponse"
/>
  <xs:complexType name="ArrayOfTracedCustomer">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="TracedCustomer" nillable="true"

```

```

type="tns:TracedCustomer" />
</xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfTracedCustomer" nillable="true" type="tns:ArrayOfTracedCustomer" />
<xs:complexType name="TracedCustomer">
  <xs:sequence>
    <xs:element name="_x003C_AccountNumber_x003E_k__BackingField" nillable="true" type="xs:string"
  />
    <xs:element name="_x003C_CustomerAddr_x003E_k__BackingField" nillable="true" type="xs:string" />
    <xs:element name="_x003C_CustomerKey_x003E_k__BackingField" nillable="true" type="xs:string" />
    <xs:element name="_x003C_CustomerName_x003E_k__BackingField" nillable="true" type="xs:string" />
    <xs:element name="_x003C_IsCritical_x003E_k__BackingField" type="xs:boolean" />
    <xs:element name="_x003C_Phone_x003E_k__BackingField" nillable="true" type="xs:string" />
    <xs:element name="_x003C_ServicePoint_x003E_k__BackingField" nillable="true" type="xs:string" />
    <xs:element name="_x003C_StationName_x003E_k__BackingField" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="TracedCustomer" nillable="true" type="tns:TracedCustomer" />
<xs:simpleType name="NetResponseErrorEnum">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NoError" />
    <xs:enumeration value="SuccessConfirm" />
    <xs:enumeration value="SystemError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
1</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="PermissionError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
2</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="RequestTimeOut">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
3</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergePredictIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
100</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
  </xs:restriction>
</xs:simpleType>

```

```

    </xs:enumeration>
    <xs:enumeration value="CanNotMoveUpNonOutageConfirmerIncidents">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
101</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="CanNotMoveUpIncidentNoProtectiveDevice">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
102</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailToGetDownstreamIncidents">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
103</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateDamageAssessmentIncidentNotFound">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
104</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateDamageAssessmentDalNotFound">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
105</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateCustomerIncidentNotFound">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
106</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateCustomerIncidentIsNonOutage">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-

```

```

107</EnumerationValue>
    </xs:appinfo>
    </xs:annotation>
  </xs:enumeration>
  <xs:enumeration value="FailedToSetPredictedStaticOnDeviceInvalidArguments">
    <xs:annotation>
      <xs:appinfo>
        <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
108</EnumerationValue>
        </xs:appinfo>
        </xs:annotation>
      </xs:enumeration>
      <xs:enumeration value="FailedToSetPredictedStaticOnDeviceNotProtective">
        <xs:annotation>
          <xs:appinfo>
            <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
109</EnumerationValue>
            </xs:appinfo>
            </xs:annotation>
          </xs:enumeration>
          <xs:enumeration value="FailedToSetPredictedStaticOnDeviceDeenergized">
            <xs:annotation>
              <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
110</EnumerationValue>
                </xs:appinfo>
                </xs:annotation>
              </xs:enumeration>
              <xs:enumeration value="FailedToSetPredictedStaticOnDeviceNoAffectedCustomers">
                <xs:annotation>
                  <xs:appinfo>
                    <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
111</EnumerationValue>
                    </xs:appinfo>
                    </xs:annotation>
                  </xs:enumeration>
                  <xs:enumeration value="FailedToSetPredictedStaticOnDeviceValidMeasurement">
                    <xs:annotation>
                      <xs:appinfo>
                        <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
112</EnumerationValue>
                        </xs:appinfo>
                        </xs:annotation>
                      </xs:enumeration>
                      <xs:enumeration value="CanNotMoveDownIncidentAsNoCallOrAMout">
                        <xs:annotation>
                          <xs:appinfo>
                            <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
113</EnumerationValue>
                            </xs:appinfo>
                            </xs:annotation>
                          </xs:enumeration>
                          <xs:enumeration value="CanNotFindProvidedDevice">

```

```

        <xs:annotation>
          <xs:appinfo>
            <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
114</EnumerationValue>
          </xs:appinfo>
        </xs:annotation>
      </xs:enumeration>
    </xs:restriction>
  </xs:simpleType>
  <xs:element name="NetResponseErrorEnum" nillable="true" type="tns:NetResponseErrorEnum" />
</xs:schema>

```

2.1.2.11. GetCustomFieldConfiguration

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.datacontract.org/2004/07/OmsInterfaceServer" />
  <xs:complexType name="ObjectCustomFieldConfiguration">
    <xs:sequence>
      <xs:element minOccurs="0" name="ObjectType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Fields" nillable="true"
xmlns:q1="http://schemas.datacontract.org/2004/07/OmsInterfaceServer"
type="q1:ArrayOfCustomFieldDescription" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ObjectCustomFieldConfiguration" nillable="true"
type="tns:ObjectCustomFieldConfiguration" />
</xs:schema>

```

2.1.2.12. GetCustomFieldConfigurationAll

The response schema is ObjectCustomFieldConfiguration. For more details, see [GetCustomFieldConfiguration](#).

2.1.2.13. GetDamage

```

<xs:schema xmlns:tns="http://schemas.datacontract.org/2004/07/OmsInterfaceServer.Models"
elementFormDefault="qualified"
targetNamespace="http://schemas.datacontract.org/2004/07/OmsInterfaceServer.Models"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="Damage">
    <xs:sequence>
      <xs:element minOccurs="0" name="AssetSubType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AssetType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DamageAssessmentLocationId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DamageId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="IncidentId" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>

```

```

    <xs:element minOccurs="0" name="IncidentName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OrderId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Quantity" type="xs:int" />
  </xs:sequence>
</xs:complexType>
<xs:element name="Damage" nillable="true" type="tns:Damage" />
</xs:schema>

```

2.1.2.14. GetDamages

```

<xs:schema xmlns:tns="http://schemas.datacontract.org/2004/07/OmsInterfaceServer.Models"
  elementFormDefault="qualified"
  targetNamespace="http://schemas.datacontract.org/2004/07/OmsInterfaceServer.Models"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="ArrayOfDamage">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="Damage" nillable="true" type="tns:Damage" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfDamage" nillable="true" type="tns:ArrayOfDamage" />
  <xs:complexType name="Damage">
    <xs:sequence>
      <xs:element minOccurs="0" name="AssetSubType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AssetType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DamageAssessmentLocationId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DamageId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="IncidentId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="IncidentName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OrderId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Quantity" type="xs:int" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="Damage" nillable="true" type="tns:Damage" />
</xs:schema>

```

2.1.2.15. GetDownStreamIncidentList

```

<xs:schema
  xmlns:tns="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
  elementFormDefault="qualified"
  targetNamespace="http://grid.energy.ge.com/service/networkoperation/networkincidentmanagementservice"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="RequestListResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="out" nillable="true" type="tns:ListOut" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:complexType>
<xs:element name="RequestListResponse" nillable="true" type="tns:RequestListResponse" />
<xs:complexType name="ListOut">
  <xs:sequence>
    <xs:element minOccurs="0" name="mRID" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ListOut" nillable="true" type="tns:ListOut" />
</xs:schema>

```

2.1.2.16. GetDownstreamIncidents

The response schema is RequestListResponse. For more details, see [GetDownStreamIncidentList](#).

2.1.2.17. GetDownstreamOrderList

The response schema is RequestListResponse. For more details, see [GetDownStreamIncidentList](#).

2.1.2.18. GetDownStreamOrderListFromDevice

The response schema is RequestListResponse. For more details, see [GetDownStreamIncidentList](#).

2.1.2.19. GetIncidentByName

```

<xs:schema xmlns:tns="http://www.esca.com/idms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="IncidentData">
    <xs:sequence>
      <xs:element name="Id" nillable="true" type="xs:string" />
      <xs:element name="Name" nillable="true" type="xs:string" />
      <xs:element name="Type" xmlns:q1="http://schemas.datacontract.org/2004/07/NetUtil"
type="q1:IncidentType" />
      <xs:element name="Status" xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil"
type="q2:eIncidentWorkingStatus" />
      <xs:element minOccurs="0" name="Feeder" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Station" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ServiceCenter" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Reason" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OpenDeviceId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OpenDeviceName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OutageDeviceType" type="xs:int" />
      <xs:element minOccurs="0" name="PhaseOut"
xmlns:q3="http://schemas.datacontract.org/2004/07/NetUtil" type="q3:ePhaseOut" />
    </xs:sequence>
  </xs:complexType>

```

```

<xs:element minOccurs="0" name="StartTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="ETR" type="xs:dateTime" />
<xs:element minOccurs="0" name="EndTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="FirstCallTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="IncidentPtX" type="xs:float" />
<xs:element minOccurs="0" name="IncidentPtY" type="xs:float" />
<xs:element minOccurs="0" name="CurrentCustomer" type="xs:int" />
<xs:element minOccurs="0" name="CriticalCustomer" type="xs:int" />
<xs:element minOccurs="0" name="CallCount" type="xs:int" />
<xs:element minOccurs="0" name="AMICount" type="xs:int" />
<xs:element minOccurs="0" name="PriorityCalls" type="xs:int" />
<xs:element minOccurs="0" name="CrewComment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewType" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewPtX" type="xs:float" />
<xs:element minOccurs="0" name="CrewPtY" type="xs:float" />
<xs:element minOccurs="0" name="TotalAffectedCustomerCount" type="xs:int" />
<xs:element minOccurs="0" name="ISDEVICEOFRECORDOPEN" type="xs:boolean" />
<xs:element minOccurs="0" name="ITR" type="xs:dateTime" />
<xs:element minOccurs="0" name="WorkOrderId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CrewStatus" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="LockFlag" type="xs:boolean" />
<xs:element minOccurs="0" name="LockUserName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OpenDevicePhaseInd" type="xs:boolean" />
<xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="FailedEquipment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OutageType" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Material" type="xs:int" />
<xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
<xs:element minOccurs="0" name="SuppressReason" type="xs:int" />
<xs:element minOccurs="0" name="OpenSwitchType" type="xs:int" />
<xs:element minOccurs="0" name="RestorationPhaseInfo" nillable="true" type="tns:RestorationData"
/>
<xs:element minOccurs="0" name="SystemAffected" type="xs:int" />
<xs:element minOccurs="0" name="IsSingleCustomer" type="xs:boolean" />
<xs:element minOccurs="0" name="IsNonOutage" type="xs:boolean" />
<xs:element minOccurs="0" name="ServiceCenterName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="LineCrewNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="TreeCrewNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="FirstResponderNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="EvaluatorNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="NoOfPolesDown" type="xs:int" />
<xs:element minOccurs="0" name="NoOfCrossArms" type="xs:int" />
<xs:element minOccurs="0" name="NoOfTreeTrimSpans" type="xs:int" />
<xs:element minOccurs="0" name="NoOfTransformers" type="xs:int" />
<xs:element minOccurs="0" name="ConductorSpan" type="xs:int" />
<xs:element minOccurs="0" name="NoServicesDown" type="xs:int" />
<xs:element minOccurs="0" name="DistributionLocationNumber" type="xs:int" />
<xs:element minOccurs="0" name="PointOfFailure" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OrgLevel1To8" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NonOutageType" type="xs:int" />
<xs:element minOccurs="0" name="CallType" type="xs:int" />

```

```

    <xs:element minOccurs="0" name="CallCountPerPhase" nillable="true"
xmlns:q4="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
type="q4:ArrayOfKeyValueOfstringint" />
    <xs:element minOccurs="0" name="MidpointSwitching" type="xs:boolean" />
    <xs:element minOccurs="0" name="Inaccessible" type="xs:boolean" />
    <xs:element minOccurs="0" name="Weather" type="xs:int" />
    <xs:element minOccurs="0" name="IsPredictedStable" type="xs:boolean" />
    <xs:element minOccurs="0" name="OriginalDeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OriginalDeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CMI" type="xs:float" />
    <xs:element minOccurs="0" name="Priority" type="xs:int" />
    <xs:element minOccurs="0" name="SelfDefinedStatus" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ETRSequenceNumber" type="xs:int" />
    <xs:element minOccurs="0" name="CaseNote" type="xs:int" />
    <xs:element minOccurs="0" name="AreaNote" nillable="true" type="tns:AreaNote" />
    <xs:element minOccurs="0" name="DeviceLocation" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IncidentSymbol" type="xs:int" />
    <xs:element minOccurs="0" name="FirstCallCustomerAddress" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="HasHazardousCustomer" type="xs:boolean" />
    <xs:element minOccurs="0" name="AMIInCount" type="xs:int" />
    <xs:element minOccurs="0" name="WorkQueues" type="xs:int" />
    <xs:element minOccurs="0" name="ActionTaken" type="xs:int" />
    <xs:element minOccurs="0" name="PowerQualityType" type="xs:int" />
    <xs:element minOccurs="0" name="FailureMode" type="xs:int" />
    <xs:element minOccurs="0" name="Manufacturer" type="xs:int" />
    <xs:element minOccurs="0" name="PublicInflictedDamageIndicator" type="xs:boolean" />
    <xs:element minOccurs="0" name="WastedTrip" type="xs:boolean" />
    <xs:element minOccurs="0" name="TroubleDescriptionsTranslated" nillable="true" type="xs:string"
/>
    <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:element name="IncidentData" nillable="true" type="tns:IncidentData" />
<xs:complexType name="RestorationData">
    <xs:sequence>
        <xs:element name="constructPhases" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="switchMask" type="xs:int" />
        <xs:element minOccurs="0" name="outageRecordLowestAor" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="deviceType" type="xs:int" />
        <xs:element minOccurs="0" name="properties" nillable="true"
xmlns:q5="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
type="q5:ArrayOfKeyValueOfstringstring" />
        <xs:element minOccurs="0" name="predictedPhases" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="predictionBlockedPhases" nillable="true" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:element name="RestorationData" nillable="true" type="tns:RestorationData" />
<xs:complexType name="AreaNote">
    <xs:sequence>
        <xs:element minOccurs="0" name="AreaNoteCode" type="xs:int" />
        <xs:element minOccurs="0" name="AreaNoteId" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="AreaNoteStartTime" type="xs:dateTime" />
        <xs:element minOccurs="0" name="AreaNoteETR" type="xs:dateTime" />
    </xs:sequence>

```

```

<xs:element minOccurs="0" name="StormName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="LevelName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SystemMode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ITRDisable" type="xs:boolean" />
<xs:element minOccurs="0" name="CreationTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="OverrideCaseNote" type="xs:boolean" />
<xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
<xs:element minOccurs="0" name="SuppressReasonCode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ParentID" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UseParent" type="xs:boolean" />
<xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Remarks" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ExcludeCMI" type="xs:boolean" />
<xs:element minOccurs="0" name="PSTDelta" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="PSTDeltaKey" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ExpirationTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="DisableAutoRestorationPings" type="xs:boolean" />
<xs:element minOccurs="0" name="ReferenceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UseFRP" type="xs:boolean" />
<xs:element minOccurs="0" name="RestorationUrgency" type="xs:boolean" />
<xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:element name="AreaNote" nillable="true" type="tns:AreaNote" />
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="tns:PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfPropertyValue" nillable="true" type="tns:ArrayOfPropertyValue" />
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element name="PropertyName" nillable="true" type="xs:string" />
    <xs:element name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="PropertyValue" nillable="true" type="tns:PropertyValue" />
</xs:schema>

```

2.1.2.20. GetIncidentsByIds

```

<xs:schema xmlns:tns="http://www.esca.com/idsm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="ArrayOfIncidentData">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="IncidentData" nillable="true"
type="tns:IncidentData" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:complexType>
<xs:element name="ArrayOfIncidentData" nillable="true" type="tns:ArrayOfIncidentData" />
<xs:complexType name="IncidentData">
  <xs:sequence>
    <xs:element name="Id" nillable="true" type="xs:string" />
    <xs:element name="Name" nillable="true" type="xs:string" />
    <xs:element name="Type" xmlns:q1="http://schemas.datacontract.org/2004/07/NetUtil"
type="q1:IncidentType" />
    <xs:element name="Status" xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil"
type="q2:eIncidentWorkingStatus" />
    <xs:element minOccurs="0" name="Feeder" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Station" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ServiceCenter" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Reason" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OpenDeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OpenDeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OutageDeviceType" type="xs:int" />
    <xs:element minOccurs="0" name="PhaseOut"
xmlns:q3="http://schemas.datacontract.org/2004/07/NetUtil" type="q3:ePhaseOut" />
    <xs:element minOccurs="0" name="StartTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="ETR" type="xs:dateTime" />
    <xs:element minOccurs="0" name="EndTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="FirstCallTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="IncidentPtX" type="xs:float" />
    <xs:element minOccurs="0" name="IncidentPtY" type="xs:float" />
    <xs:element minOccurs="0" name="CurrentCustomer" type="xs:int" />
    <xs:element minOccurs="0" name="CriticalCustomer" type="xs:int" />
    <xs:element minOccurs="0" name="CallCount" type="xs:int" />
    <xs:element minOccurs="0" name="AMICount" type="xs:int" />
    <xs:element minOccurs="0" name="PriorityCalls" type="xs:int" />
    <xs:element minOccurs="0" name="CrewComment" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewPtX" type="xs:float" />
    <xs:element minOccurs="0" name="CrewPtY" type="xs:float" />
    <xs:element minOccurs="0" name="TotalAffectedCustomerCount" type="xs:int" />
    <xs:element minOccurs="0" name="ISDEVICEOFRECORDOPEN" type="xs:boolean" />
    <xs:element minOccurs="0" name="ITR" type="xs:dateTime" />
    <xs:element minOccurs="0" name="WorkOrderId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CrewStatus" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="LockFlag" type="xs:boolean" />
    <xs:element minOccurs="0" name="LockUserName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OpenDevicePhaseInd" type="xs:boolean" />
    <xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="FailedEquipment" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OutageType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Material" type="xs:int" />
    <xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
    <xs:element minOccurs="0" name="SuppressReason" type="xs:int" />
    <xs:element minOccurs="0" name="OpenSwitchType" type="xs:int" />
    <xs:element minOccurs="0" name="RestorationPhaseInfo" nillable="true" type="tns:RestorationData"

```

```

/>
  <xs:element minOccurs="0" name="SystemAffected" type="xs:int" />
  <xs:element minOccurs="0" name="IsSingleCustomer" type="xs:boolean" />
  <xs:element minOccurs="0" name="IsNonOutage" type="xs:boolean" />
  <xs:element minOccurs="0" name="ServiceCenterName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="LineCrewNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="TreeCrewNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="FirstResponderNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="EvaluatorNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="NoOfPolesDown" type="xs:int" />
  <xs:element minOccurs="0" name="NoOfCrossArms" type="xs:int" />
  <xs:element minOccurs="0" name="NoOfTreeTrimSpans" type="xs:int" />
  <xs:element minOccurs="0" name="NoOfTransformers" type="xs:int" />
  <xs:element minOccurs="0" name="ConductorSpan" type="xs:int" />
  <xs:element minOccurs="0" name="NoServicesDown" type="xs:int" />
  <xs:element minOccurs="0" name="DistributionLocationNumber" type="xs:int" />
  <xs:element minOccurs="0" name="PointOfFailure" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="OrgLevel1To8" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="NonOutageType" type="xs:int" />
  <xs:element minOccurs="0" name="CallType" type="xs:int" />
  <xs:element minOccurs="0" name="CallCountPerPhase" nillable="true"
xmlns:q4="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
type="q4:ArrayOfKeyValueOfstringint" />
  <xs:element minOccurs="0" name="MidpointSwitching" type="xs:boolean" />
  <xs:element minOccurs="0" name="Inaccessible" type="xs:boolean" />
  <xs:element minOccurs="0" name="Weather" type="xs:int" />
  <xs:element minOccurs="0" name="IsPredictedStable" type="xs:boolean" />
  <xs:element minOccurs="0" name="OriginalDeviceId" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="OriginalDeviceName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="CMI" type="xs:float" />
  <xs:element minOccurs="0" name="Priority" type="xs:int" />
  <xs:element minOccurs="0" name="SelfDefinedStatus" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="ETRSequenceNumber" type="xs:int" />
  <xs:element minOccurs="0" name="CaseNote" type="xs:int" />
  <xs:element minOccurs="0" name="AreaNote" nillable="true" type="tns:AreaNote" />
  <xs:element minOccurs="0" name="DeviceLocation" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="IncidentSymbol" type="xs:int" />
  <xs:element minOccurs="0" name="FirstCallCustomerAddress" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="HasHazardousCustomer" type="xs:boolean" />
  <xs:element minOccurs="0" name="AMIInCount" type="xs:int" />
  <xs:element minOccurs="0" name="WorkQueues" type="xs:int" />
  <xs:element minOccurs="0" name="ActionTaken" type="xs:int" />
  <xs:element minOccurs="0" name="PowerQualityType" type="xs:int" />
  <xs:element minOccurs="0" name="FailureMode" type="xs:int" />
  <xs:element minOccurs="0" name="Manufacturer" type="xs:int" />
  <xs:element minOccurs="0" name="PublicInflictedDamageIndicator" type="xs:boolean" />
  <xs:element minOccurs="0" name="WastedTrip" type="xs:boolean" />
  <xs:element minOccurs="0" name="TroubleDescriptionsTranslated" nillable="true" type="xs:string"
/>
  <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:element name="IncidentData" nillable="true" type="tns:IncidentData" />

```

```

<xs:complexType name="RestorationData">
  <xs:sequence>
    <xs:element name="constructPhases" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="switchMask" type="xs:int" />
    <xs:element minOccurs="0" name="outageRecordLowestAor" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="deviceType" type="xs:int" />
    <xs:element minOccurs="0" name="properties" nillable="true"
xmlns:q5="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
type="q5:ArrayOfKeyValueOfstringstring" />
    <xs:element minOccurs="0" name="predictedPhases" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="predictionBlockedPhases" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="RestorationData" nillable="true" type="tns:RestorationData" />
<xs:complexType name="AreaNote">
  <xs:sequence>
    <xs:element minOccurs="0" name="AreaNoteCode" type="xs:int" />
    <xs:element minOccurs="0" name="AreaNoteId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="AreaNoteStartTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="AreaNoteETR" type="xs:dateTime" />
    <xs:element minOccurs="0" name="StormName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="LevelName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="SystemMode" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ITRDisable" type="xs:boolean" />
    <xs:element minOccurs="0" name="CreationTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="OverrideCaseNote" type="xs:boolean" />
    <xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
    <xs:element minOccurs="0" name="SuppressReasonCode" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ParentID" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="UseParent" type="xs:boolean" />
    <xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Remarks" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ExcludeCMI" type="xs:boolean" />
    <xs:element minOccurs="0" name="PSTDelta" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="PSTDeltaKey" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ExpirationTime" type="xs:dateTime" />
    <xs:element minOccurs="0" name="DisableAutoRestorationPings" type="xs:boolean" />
    <xs:element minOccurs="0" name="ReferenceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="UseFRP" type="xs:boolean" />
    <xs:element minOccurs="0" name="RestorationUrgency" type="xs:boolean" />
    <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:element name="AreaNote" nillable="true" type="tns:AreaNote" />
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="tns:PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfPropertyValue" nillable="true" type="tns:ArrayOfPropertyValue" />
<xs:complexType name="PropertyValue">
  <xs:sequence>

```

```

    <xs:element name="PropertyName" nillable="true" type="xs:string" />
    <xs:element name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="PropertyValue" nillable="true" type="tns:PropertyValue" />
</xs:schema>

```

2.1.2.21. GetOrdersOnFeeder

The response schema is RequestListResponse. For more details, see [GetDownStreamIncidentList](#).

2.1.2.22. GetStructureById

```

<xs:schema xmlns:tns="http://schemas.datacontract.org/2004/07/OmsInterfaceServer.Models"
  elementFormDefault="qualified"
  targetNamespace="http://schemas.datacontract.org/2004/07/OmsInterfaceServer.Models"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="Structure">
    <xs:sequence>
      <xs:element minOccurs="0" name="Height" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OutageDeviceType" type="xs:int" />
      <xs:element minOccurs="0" name="Owner" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StructureId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StructureName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StructureType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StructureTypeIndex" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="X" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Y" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="Structure" nillable="true" type="tns:Structure" />
</xs:schema>

```

2.1.2.23. GetWorkOrdersForIncident

```

<xs:schema xmlns:tns="http://schemas.datacontract.org/2004/07/NetUtil" elementFormDefault="qualified"
  targetNamespace="http://schemas.datacontract.org/2004/07/NetUtil"
  xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/" />
  <xs:complexType name="GetWorkOrdersForIncidentResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="ActiveOrderId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="RelatedOrderIds" nillable="true"
        xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
      <xs:element minOccurs="0" name="Status" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="StatusCode" type="tns:NetResponseErrorEnum" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:complexType>
<xs:element name="GetWorkOrdersForIncidentResponse" nillable="true"
type="tns:GetWorkOrdersForIncidentResponse" />
<xs:simpleType name="NetResponseErrorEnum">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NoError" />
    <xs:enumeration value="SuccessConfirm" />
    <xs:enumeration value="SystemError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
1</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="PermissionError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
2</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="RequestTimeOut">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
3</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergePredictIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
100</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="CanNotMoveUpNonOutageConfirmerIncidents">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
101</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="CanNotMoveUpIncidentNoProtectiveDevice">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
102</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
  </xs:restriction>
</xs:simpleType>

```

```

        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailToGetDownstreamIncidents">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
103</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateDamageAssessmentIncidentNotFound">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
104</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateDamageAssessmentDalNotFound">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
105</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateCustomerIncidentNotFound">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
106</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToAssociateCustomerIncidentIsNonOutage">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
107</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToSetPredictedStaticOnDeviceInvalidArguments">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
108</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToSetPredictedStaticOnDeviceNotProtective">
        <xs:annotation>
            <xs:appinfo>

```

```

        <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
109</EnumerationValue>
        </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="FailedToSetPredictedStaticOnDeviceDeenergized">
        <xs:annotation>
        <xs:appinfo>
            <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
110</EnumerationValue>
            </xs:appinfo>
            </xs:annotation>
        </xs:enumeration>
        <xs:enumeration value="FailedToSetPredictedStaticOnDeviceNoAffectedCustomers">
            <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
111</EnumerationValue>
                </xs:appinfo>
                </xs:annotation>
            </xs:enumeration>
            <xs:enumeration value="FailedToSetPredictedStaticOnDeviceValidMeasurement">
                <xs:annotation>
                <xs:appinfo>
                    <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
112</EnumerationValue>
                    </xs:appinfo>
                    </xs:annotation>
                </xs:enumeration>
                <xs:enumeration value="CanNotMoveDownIncidentAsNoCallOrAMout">
                    <xs:annotation>
                    <xs:appinfo>
                        <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
113</EnumerationValue>
                        </xs:appinfo>
                        </xs:annotation>
                    </xs:enumeration>
                    <xs:enumeration value="CanNotFindProvidedDevice">
                        <xs:annotation>
                        <xs:appinfo>
                            <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
114</EnumerationValue>
                            </xs:appinfo>
                            </xs:annotation>
                        </xs:enumeration>
                        <xs:enumeration value="IncidentNotFound">
                            <xs:annotation>
                            <xs:appinfo>
                                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
115</EnumerationValue>
                                </xs:appinfo>
                                </xs:annotation>
                            </xs:enumeration>

```

```

    <xs:enumeration value="FailedToAssociateCustomerIncidentIsAlreadyMaster">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
116</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergeIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
117</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergeOutageToNonoutageIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
118</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergeMomentaryIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
119</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergePlannedIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
120</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergePendingClearIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
121</EnumerationValue>
        </xs:appinfo>
      </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergeToSingleCustomerIncidentError">
      <xs:annotation>
        <xs:appinfo>
          <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
122</EnumerationValue>

```

```

        </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
    <xs:enumeration value="MergeToSameIncidentError">
        <xs:annotation>
            <xs:appinfo>
                <EnumerationValue xmlns="http://schemas.microsoft.com/2003/10/Serialization/">-
123</EnumerationValue>
            </xs:appinfo>
        </xs:annotation>
    </xs:enumeration>
</xs:restriction>
</xs:simpleType>
<xs:element name="NetResponseErrorEnum" nillable="true" type="tns:NetResponseErrorEnum" />
</xs:schema>

```

2.1.2.24. Health

The response schema is RequestResponse. For more details, see [DeleteAreaNote](#).

2.1.2.25. Incidents

```

<xs:schema xmlns:tns="http://www.esca.com/idsms" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/idsms" xmlns:xs="http://www.w3.org/2001/XMLSchema">
    <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
    <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
    <xs:complexType name="ArrayOfIncidentData">
        <xs:sequence>
            <xs:element minOccurs="0" maxOccurs="unbounded" name="IncidentData" nillable="true"
type="tns:IncidentData" />
        </xs:sequence>
    </xs:complexType>
    <xs:element name="ArrayOfIncidentData" nillable="true" type="tns:ArrayOfIncidentData" />
    <xs:complexType name="IncidentData">
        <xs:sequence>
            <xs:element name="Id" nillable="true" type="xs:string" />
            <xs:element name="Name" nillable="true" type="xs:string" />
            <xs:element name="Type" xmlns:q1="http://schemas.datacontract.org/2004/07/NetUtil"
type="q1:IncidentType" />
            <xs:element name="Status" xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil"
type="q2:eIncidentWorkingStatus" />
            <xs:element minOccurs="0" name="Feeder" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="Station" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="ServiceCenter" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="Reason" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="OpenDeviceId" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="OpenDeviceName" nillable="true" type="xs:string" />
            <xs:element minOccurs="0" name="OutageDeviceType" type="xs:int" />
            <xs:element minOccurs="0" name="PhaseOut"

```

```

xmlns:q3="http://schemas.datacontract.org/2004/07/NetUtil" type="q3:ePhaseOut" />
  <xs:element minOccurs="0" name="StartTime" type="xs:dateTime" />
  <xs:element minOccurs="0" name="ETR" type="xs:dateTime" />
  <xs:element minOccurs="0" name="EndTime" type="xs:dateTime" />
  <xs:element minOccurs="0" name="FirstCallTime" type="xs:dateTime" />
  <xs:element minOccurs="0" name="IncidentPtX" type="xs:float" />
  <xs:element minOccurs="0" name="IncidentPtY" type="xs:float" />
  <xs:element minOccurs="0" name="CurrentCustomer" type="xs:int" />
  <xs:element minOccurs="0" name="CriticalCustomer" type="xs:int" />
  <xs:element minOccurs="0" name="CallCount" type="xs:int" />
  <xs:element minOccurs="0" name="AMICount" type="xs:int" />
  <xs:element minOccurs="0" name="PriorityCalls" type="xs:int" />
  <xs:element minOccurs="0" name="CrewComment" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="CrewId" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="CrewName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="CrewType" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="CrewPtX" type="xs:float" />
  <xs:element minOccurs="0" name="CrewPtY" type="xs:float" />
  <xs:element minOccurs="0" name="TotalAffectedCustomerCount" type="xs:int" />
  <xs:element minOccurs="0" name="ISDEVICEOFRECORDOPEN" type="xs:boolean" />
  <xs:element minOccurs="0" name="ITR" type="xs:dateTime" />
  <xs:element minOccurs="0" name="WorkOrderId" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="CrewStatus" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="LockFlag" type="xs:boolean" />
  <xs:element minOccurs="0" name="LockUserName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="OpenDevicePhaseInd" type="xs:boolean" />
  <xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="FailedEquipment" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="OutageType" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="Material" type="xs:int" />
  <xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
  <xs:element minOccurs="0" name="SuppressReason" type="xs:int" />
  <xs:element minOccurs="0" name="OpenSwitchType" type="xs:int" />
  <xs:element minOccurs="0" name="RestorationPhaseInfo" nillable="true" type="tns:RestorationData"
/>
  <xs:element minOccurs="0" name="SystemAffected" type="xs:int" />
  <xs:element minOccurs="0" name="IsSingleCustomer" type="xs:boolean" />
  <xs:element minOccurs="0" name="IsNonOutage" type="xs:boolean" />
  <xs:element minOccurs="0" name="ServiceCenterName" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="LineCrewNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="TreeCrewNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="FirstResponderNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="EvaluatorNeeded" type="xs:boolean" />
  <xs:element minOccurs="0" name="NoOfPolesDown" type="xs:int" />
  <xs:element minOccurs="0" name="NoOfCrossArms" type="xs:int" />
  <xs:element minOccurs="0" name="NoOfTreeTrimSpans" type="xs:int" />
  <xs:element minOccurs="0" name="NoOfTransformers" type="xs:int" />
  <xs:element minOccurs="0" name="ConductorSpan" type="xs:int" />
  <xs:element minOccurs="0" name="NoServicesDown" type="xs:int" />
  <xs:element minOccurs="0" name="DistributionLocationNumber" type="xs:int" />
  <xs:element minOccurs="0" name="PointOfFailure" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="OrgLevel1To8" nillable="true" type="xs:string" />
  <xs:element minOccurs="0" name="NonOutageType" type="xs:int" />

```

```

    <xs:element minOccurs="0" name="CallType" type="xs:int" />
    <xs:element minOccurs="0" name="CallCountPerPhase" nillable="true"
xmlns:q4="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
type="q4:ArrayOfKeyValueOfstringint" />
    <xs:element minOccurs="0" name="MidpointSwitching" type="xs:boolean" />
    <xs:element minOccurs="0" name="Inaccessible" type="xs:boolean" />
    <xs:element minOccurs="0" name="Weather" type="xs:int" />
    <xs:element minOccurs="0" name="IsPredictedStable" type="xs:boolean" />
    <xs:element minOccurs="0" name="OriginalDeviceId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="OriginalDeviceName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="CMI" type="xs:float" />
    <xs:element minOccurs="0" name="Priority" type="xs:int" />
    <xs:element minOccurs="0" name="SelfDefinedStatus" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="ETRSequenceNumber" type="xs:int" />
    <xs:element minOccurs="0" name="CaseNote" type="xs:int" />
    <xs:element minOccurs="0" name="AreaNote" nillable="true" type="tns:AreaNote" />
    <xs:element minOccurs="0" name="DeviceLocation" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="IncidentSymbol" type="xs:int" />
    <xs:element minOccurs="0" name="FirstCallCustomerAddress" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="HasHazardousCustomer" type="xs:boolean" />
    <xs:element minOccurs="0" name="AMIInCount" type="xs:int" />
    <xs:element minOccurs="0" name="WorkQueues" type="xs:int" />
    <xs:element minOccurs="0" name="ActionTaken" type="xs:int" />
    <xs:element minOccurs="0" name="PowerQualityType" type="xs:int" />
    <xs:element minOccurs="0" name="FailureMode" type="xs:int" />
    <xs:element minOccurs="0" name="Manufacturer" type="xs:int" />
    <xs:element minOccurs="0" name="PublicInflictedDamageIndicator" type="xs:boolean" />
    <xs:element minOccurs="0" name="WastedTrip" type="xs:boolean" />
    <xs:element minOccurs="0" name="TroubleDescriptionsTranslated" nillable="true" type="xs:string"
/>
    <xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:element name="IncidentData" nillable="true" type="tns:IncidentData" />
<xs:complexType name="RestorationData">
    <xs:sequence>
        <xs:element name="constructPhases" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="switchMask" type="xs:int" />
        <xs:element minOccurs="0" name="outageRecordLowestAor" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="deviceType" type="xs:int" />
        <xs:element minOccurs="0" name="properties" nillable="true"
xmlns:q5="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
type="q5:ArrayOfKeyValueOfstringstring" />
        <xs:element minOccurs="0" name="predictedPhases" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="predictionBlockedPhases" nillable="true" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:element name="RestorationData" nillable="true" type="tns:RestorationData" />
<xs:complexType name="AreaNote">
    <xs:sequence>
        <xs:element minOccurs="0" name="AreaNoteCode" type="xs:int" />
        <xs:element minOccurs="0" name="AreaNoteId" nillable="true" type="xs:string" />
        <xs:element minOccurs="0" name="AreaNoteStartTime" type="xs:dateTime" />
    </xs:sequence>

```

```

<xs:element minOccurs="0" name="AreaNoteETR" type="xs:dateTime" />
<xs:element minOccurs="0" name="StormName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="LevelName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="SystemMode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ITRDisable" type="xs:boolean" />
<xs:element minOccurs="0" name="CreationTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="OverrideCaseNote" type="xs:boolean" />
<xs:element minOccurs="0" name="SuppressNotification" type="xs:boolean" />
<xs:element minOccurs="0" name="SuppressReasonCode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ParentID" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UseParent" type="xs:boolean" />
<xs:element minOccurs="0" name="AOR" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Remarks" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ExcludeCMI" type="xs:boolean" />
<xs:element minOccurs="0" name="PSTDelta" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="PSTDeltaKey" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ExpirationTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="DisableAutoRestorationPings" type="xs:boolean" />
<xs:element minOccurs="0" name="ReferenceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="UseFRP" type="xs:boolean" />
<xs:element minOccurs="0" name="RestorationUrgency" type="xs:boolean" />
<xs:element minOccurs="0" name="CustomFields" nillable="true" type="tns:ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:element name="AreaNote" nillable="true" type="tns:AreaNote" />
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="tns:PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfPropertyValue" nillable="true" type="tns:ArrayOfPropertyValue" />
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element name="PropertyName" nillable="true" type="xs:string" />
    <xs:element name="Value" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="PropertyValue" nillable="true" type="tns:PropertyValue" />
</xs:schema>

```

2.1.2.26. UpdateRestorationStep

The response schema is RequestResponse. For more details, see [DeleteAreaNote](#).

2.1.2.27. UpdateSwitchStatus

The response schema is RequestResponse. For more details, see [DeleteAreaNote](#).

2.2. Query Web Service

Note

The Query web service is a public web service, which can be used by projects to query OMS data from the OMS server.

The OMS Interface Server Query web service is configured by OMS Interface Server Web Service Plugin parameters in the `OmsInterfaceServerConfig.xml` configuration file and by OMS Interface Query Service Plugin parameters in configuration files for applications that communicate with the Query web service (for example, `OmsAdapterConfig.xml`).

The following are default OMS Interface Server plugin web service URLs:

- **/api/adms/oms/v2/**: This is the root of the OMS Interface Server web service.
 - **Incident?select=<CommaSeparatedQueryFieldNames>**: Use this message to return data for all incidents in the system. You can optionally specify incident query fields to return data for. (GET)

For example: `http://localhost:18802/api/adms/oms/v2/Incident?select=id,name`
 - **Incident?name=<Name>&select=<CommaSeparatedQueryFieldNames>**: Use this message to return data for an incident by name. (GET)
 - **Incident?id=<ID>&select=<CommaSeparatedQueryFieldNames>**: Use this message to return data for an incident by ID. (GET)
 - **CompletedIncident?name=<Name>&select=<CommaSeparatedQueryFieldNames>**: Use this message to return incident and customer data for a completed incident by name. (GET)
 - **IncidentWithCustomers?name=<Name>&select=<CommaSeparatedQueryFieldNames>**: Use this message to return incident and customer data for an incident by name. (GET)
 - **PlannedOutage?id=<ID>&select=<CommaSeparatedQueryFieldNames>**: Use this message to return data for a planned outage by ID. (GET)
 - **Oms?select=<CommaSeparatedQueryFieldNames>&from=<ObjectName>&where=<Where>**: Use this message to query OMS data from NetServer and get the result in JSON format. This message supports similar query objects as those queried by OMS grid tabular displays. (GET)

Example 1 (Query incident data):

`http://localhost:18802/api/adms/oms/v2/Oms?select=id,name&from=oms.rt.incident`

Example 2 (Query order data):

`http://localhost:18802/api/adms/oms/v2/Oms?select=id,name&from=WORKORDERLIST`

Example 3 (Query order AMI Analysis data):

`http://localhost:18802/api/adms/oms/v2/Oms?select=SYSTEMSTATUS,PINGSTATUS,TOTALAMIINCOUNT,TOTALCUSTCOUNT,METERSOUT,LASTPING,TRANSFORMERSLISTJSON,TOTALTF
COUNT&from=AMIANALYSIS(Incident_Name)`

Example 4 (Query incident chronology):

`http://localhost:18802/api/adms/oms/v2/Oms?select=time,msgtype,username,action&from=ChronologyLog(Incident_Name)`

Example 5 (Query planned outage affected customers):

`http://localhost:18802/api/adms/oms/v2/Oms?select=CustomerName,AreaCode,customerkey,&from=plannedoutage(Planned_Outage_ID).PLANNEDAFFECTEDCUSTOMERS`

- **OmsTable?select=<CommaSeparatedQueryFieldNames>&from=<ObjectName>&where=<Where>**: Use this message to query OMS data from NetServer and get the result in the tabular format. (GET)

For example:

`http://localhost:18802/api/adms/oms/v2/OmsTable?select=id,name&from=incident`

- **Dms?select=<CommaSeparatedQueryFieldNames>&from=<ObjectName>&where=<Where>**: This message queries DMS data from NetServer and gets the result in JSON format. This message is designed and used for internal purposes. To query DMS data, use DMS Interface server messages instead of this message. (GET)
- **Oms?<OmsQueryRequest>**: Use this message to post an OMS query request. (POST)
- **HierarchyGroups**: Use this message to return a hierarchy tree with all hierarchy elements, such as specific feeders and the company. (GET)
- **Regions?groupType=<ElementGroupType>&id=<ElementID>**: Use this message to return hierarchy for a system element. (GET)

For example:

`http://localhost:18802/api/adms/oms/v2/Regions?groupType=STATION&id=EQSI`

3. OMS Adapter

The OMS Adapter provides a web services portal for sending OMS notifications to web clients and communicating with the GE's Outage Assist solution and other external applications. The OMS Adapter receives OMS data from the OMS Interface server.

The OMS adapter reads call, incident, crew, and de-energized line journal files created by the OMS server and creates the OMS Adapter journal file for sending updates to external applications.

The OMS Adapter is developed using Amazon ElastiCache, which is a fully managed in-memory data store and cache service by Amazon Web Services (AWS). The OMS Adapter is built on top of the Adapter framework, which is also used to build other services, such as the View Outage Portal Adapter.

While communicating using the OMS Adapter, interface messages are exchanged via web services. ADMS initiates all message exchanges with external applications. External applications never post messages to ADMS.

Multiple instances of the OMS Adapter can run simultaneously.

When a supported event type occurs in ADMS, event data is sent to a web client via the OMS Adapter. Notifications are sent on demand using a POST message type. Both JSON and XML message formats are supported.

The OMS Adapter web interface is implemented using SOAP 1.2. OMS Data Collections are provided through the Web Services Description Language (WSDL). For details about SOAP, refer to <https://www.w3.org/TR/soap/>. For details about WSDL, refer to <http://schemas.xmlsoap.org/wsdl/>.

4. Customer Experience Adapter

The Customer Experience (CE) Adapter supports sending the following notifications to a web client.

Name	Type	Description
IncidentUpdateNotification	Output	Notification about creating or updating an incident in ADMS. Incident parameters whose update triggers this message are configurable. The notification message contains all incident properties (ID, ETR, Phase, etc.) and affected, called, and restored customers. By default, the following events trigger this notification: creating, confirming, canceling, splitting, merging, and clearing an incident; prediction change; blocking or canceling prediction; partial restoration; setting an incident to pending clear state; incident field changes, including the working status, cause, case note, action taken, ETR, ETA, ETA crew, suppression flag, and storm mode changes; crew status changes to pending assigned, assigned, enroute, and onsite. Customer apology notifications (notifications without an incident number) are sent to customers that are associated with canceled incidents or predicted incidents where the prediction was blocked or partially confirmed. Note By default, a notification about creating a momentary incident is disabled (configurable). However, a notification is sent when a momentary incident becomes confirmed.
PlannedOutageNotification	Output	Notification about creating a planned outage in ADMS.
UpdatePlannedOutageNotification	Output	Notification about updating a planned outage in ADMS. Planned outage parameters whose update triggers this message are configurable.
DeletePlannedOutageNotification	Output	Notification about removing a planned outage in ADMS.

The web service path structure is /api/CE/v2/ service root.

4.1. Elements

All Customer Experience Adapter web service elements are listed in the schema. Message schema files are provided in two formats, XSD and OpenApi. Each message type is defined using XSD and the OpenApi specification, which also defines the service paths.

For reference, the elements are listed below.

Note

In the IncidentUpdateNotification message, the CaseNoteCode and CaseNoteText values are set according to the incident's case note. The AreaSettingText value is set to the incident area setting's area note.

IncidentUpdateNotification

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="IncidentUpdateNotification" nillable="true" type="IncidentUpdateNotification" />
  <xs:complexType name="IncidentUpdateNotification">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="AffectedCustomers"
type="ArrayOfIncidentCustomer" />
          <xs:element minOccurs="0" maxOccurs="1" name="CalledCustomers"
type="ArrayOfIncidentCustomer" />
          <xs:element minOccurs="0" maxOccurs="1" name="EnergizedCustomers"
type="ArrayOfIncidentCustomer" />
          <xs:element minOccurs="0" maxOccurs="1" name="ApologyCustomers"
type="ArrayOfIncidentCustomer" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentID" nillable="true" type="xs:string"
/>
          <xs:element minOccurs="0" maxOccurs="1" name="OutageIslandIds" type="ArrayOfUnsignedInt" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentName" nillable="true" type="xs:string"
/>
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentWorkingStatusCode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentWorkingStatus" nillable="true"
type="eIncidentWorkingStatus" />
          <xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="CauseCode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CurrentCustomerCount" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceID" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FeederName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Suppressionflag" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="StormMode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="Regions" type="ArrayOfRegionLevel" />
          <xs:element minOccurs="1" maxOccurs="1" name="ETA" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="ETACrew" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="ClearTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PhaseOut" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CaseNoteCode" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteText" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AreaSettingText" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Actiontaken" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="ActionCausingUpdate" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceOpen" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentPtX" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentPtY" type="xs:float" />
          <xs:element minOccurs="0" maxOccurs="1" name="UpdateFieldName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentType" type="xs:int" />
        
```

```

        <xs:element minOccurs="0" maxOccurs="1" name="PlannedOutageId" type="xs:string" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfIncidentCustomer">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="IncidentCustomer" nillable="true"
type="IncidentCustomer" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="IncidentCustomer">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="CustIdx" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ServicePoint" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AccountNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="IncidentID" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="PhoneCall" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastCallTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="EnergizedTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="CallReferenceId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="SuspendCustomer" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="WantsCallback" type="xs:boolean" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfUnsignedInt">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="unsignedInt" type="xs:unsignedInt" />
    </xs:sequence>
</xs:complexType>

```

```

<xs:simpleType name="eIncidentWorkingStatus">
  <xs:restriction base="xs:string">
    <xs:enumeration value="Unassigned" />
    <xs:enumeration value="SentToWfm" />
    <xs:enumeration value="PendingDispatched" />
    <xs:enumeration value="Dispatched" />
    <xs:enumeration value="EnRoute" />
    <xs:enumeration value="OnSite" />
    <xs:enumeration value="Cancel" />
    <xs:enumeration value="Complete" />
    <xs:enumeration value="PendingClear" />
    <xs:enumeration value="Patrolled" />
    <xs:enumeration value="Evaluating" />
    <xs:enumeration value="Evaluated" />
    <xs:enumeration value="Restored" />
    <xs:enumeration value="Cleared" />
    <xs:enumeration value="Assigned" />
    <xs:enumeration value="MaxValue" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfRegionLevel">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="RegionLevel" nillable="true"
type="RegionLevel" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="RegionLevel">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="RegionType" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RegionName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RegionID" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

PlannedOutageNotification

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="PlannedOutageNotification" nillable="true" type="PlannedOutageNotification" />
  <xs:complexType name="PlannedOutageNotification">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="CreatedTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryEndTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="AlternateStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="AlternateEndTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="UseAlternateDate" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="EventType" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

    <xs:element minOccurs="1" maxOccurs="1" name="CausesIncident" type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="DevicePtX" type="xs:float" />
    <xs:element minOccurs="1" maxOccurs="1" name="DevicePtY" type="xs:float" />
    <xs:element minOccurs="1" maxOccurs="1" name="DeviceType" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceTypeString" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ActionCausingUpdate" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="SwitchType" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="CustomerAccountsString" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="CustomerAccountsNotifyString" type="xs:string"
  />
  <xs:element minOccurs="0" maxOccurs="1" name="AddedCustomersString" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="AddedCustomersNotifyString" type="xs:string"
  />
  <xs:element minOccurs="0" maxOccurs="1" name="RemovedCustomersString" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="RemovedCustomersNotifyString" type="xs:string"
  />
  <xs:element minOccurs="0" maxOccurs="1" name="Station" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Regions" type="ArrayOfRegionLevel" />
  <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeName" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeNumber" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeEmail" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeAddress" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="OutageMessage" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CreatorComments" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="AOR" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="SwitchOrderId" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="SwitchOrderName" type="xs:string" />
  <xs:element minOccurs="1" maxOccurs="1" name="Modified" type="xs:dateTime" />
  <xs:element minOccurs="0" maxOccurs="1" name="FeederName" type="xs:string" />
  <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
  <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
  <xs:element minOccurs="1" maxOccurs="1" name="PlannedOutageType" type="xs:int" />
  <xs:element minOccurs="0" maxOccurs="1" name="UpdateFieldName" type="xs:string" />
7   <xs:element minOccurs="0" maxOccurs="1" name="Interruptions" type="ArrayOfInterruption" />
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
7 <xs:complexType name="ArrayOfInterruption">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="Interruption" nillable="true"
type="Interruption" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="Interruption">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="InterruptionId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ScheduleStartTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="ScheduleEndTime" type="xs:dateTime" />

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="Duration" type="TimeSpan" />
        <xs:element minOccurs="1" maxOccurs="1" name="AffectedCustomersCount" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CMI" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="PlannedOutageId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AffectedCustomersList" type="ArrayOfString" />
        <xs:element minOccurs="0" maxOccurs="1" name="AffectedCustomersString" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="TimeSpan" />
<xs:complexType name="ArrayOfString">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="string" nillable="true" type="xs:string"
    />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfRegionLevel">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="RegionLevel" nillable="true"
type="RegionLevel" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="RegionLevel">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="RegionType" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RegionName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RegionID" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

DeletePlannedOutageNotification

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeletePlannedOutageNotification" nillable="true"
type="DeletePlannedOutageNotification" />
  <xs:complexType name="DeletePlannedOutageNotification">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsDeleted" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsCancelled" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

UpdatePlannedOutageNotification

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="UpdatePlannedOutageNotification" nillable="true"
type="UpdatePlannedOutageNotification" />
  <xs:complexType name="UpdatePlannedOutageNotification">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FieldName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="OldValue" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="NewValue" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="Created" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="Modified" type="xs:dateTime" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

5. ANPS Adapter

The Automated Notification Paging Services (ANPS) Adapter supports sending the following notifications (see the below table) to a web client.

Name	Type	Description
ANPSNotification	Output	Sends notifications about area setting changes and breaker outages in ADMS.

Note

Different fields appear in the notification payload for different event types (area setting changes or breaker outages).

Note

ANPS notification is not set when the following area setting fields are updated: InternalNote, DisableAutoRestorationPings, UseFrp, RestorationUrgency, PstDeltaKey, ExcludeCmi, ItrDisable, LastUserId, or LastUpdateTime.

The ANPS Adapter web service path structure is /api/ANPS/v2/ service root.

ANPSNotification Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="ANPSNotification" nillable="true" type="ANPSNotification" />
  <xs:complexType name="ANPSNotification">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentID" nillable="true" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Areatypecode" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ActionCausingUpdate" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceID" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="PhaseOut" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceOpen" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="StormMode" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="AreaId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AreaAffected" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FeederID" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FeederName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="StationName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Suppressionflag" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="SuppressionCode" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="SuppressionText" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Regions" type="ArrayOfRegionLevel" />
          <xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="0" maxOccurs="1" name="Remarks" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CauseCode" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CaseNoteCode" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteText" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CurrentCustomerCount" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CreationTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastUpdateTime" nillable="true"
type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="LastUpdateByUserId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="StormName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CustomFields" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfRegionLevel">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="RegionLevel" nillable="true"
type="RegionLevel" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="RegionLevel">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="RegionType" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RegionName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RegionID" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

6. View Outage Portal Adapter

The View Outage Portal (VOP) Adapter supports sending the following notifications to a web client.

Name	Type	Description
ViewOutagePortalNotification	Output	Sends information from the De-Energized Line Journal, which includes data about de-energized lines, predicted out lines, and cuts by substation.

The View Outage Portal Adapter web service path structure is /api/ext/ANPS/v2/ service root.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="ViewOutagePortalNotification" nillable="true" type="ViewOutagePortalNotification" />
  <xs:complexType name="ViewOutagePortalNotification">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="StationSummary" type="ArrayOfStationSummary" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true" type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfStationSummary">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="StationSummary" nillable="true" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

type="StationSummary" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="StationSummary">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="DeenergizedLines" type="ArrayOfLineSection" />
    <xs:element minOccurs="0" maxOccurs="1" name="PredictedLines" type="ArrayOfLineSection" />
    <xs:element minOccurs="0" maxOccurs="1" name="Cuts" type="ArrayOfCut" />
    <xs:element minOccurs="0" maxOccurs="1" name="StationName" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfLineSection">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="LineSection" nillable="true"
type="LineSection" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="LineSection">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="LineID" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsOverhead" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="OutageIslandId" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Vertices" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCut">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="Cut" nillable="true" type="Cut" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="Cut">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="OriginalLineID" nillable="true" type="xs:string"
/>
    <xs:element minOccurs="1" maxOccurs="1" name="LineID" nillable="true" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="OutageIslandId" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Vertices" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7. Running the OMS Interface Server and ADMS Adapters

7.1. Running the OMS Interface Server

To start the OMS Interface server, open the Windows Start menu and navigate to All Programs> Eterra> Distribution> ADMS Interfaces> OMS Interface Server.

Note

The OMS Interface server must be run by a user with administrator privileges.

By default, the OMS Interface server starts with the OmsInterfaceServerConfig.xml configuration file. For more details about the OMS Interface server configuration, see [OMS Interface Server Configuration](#).

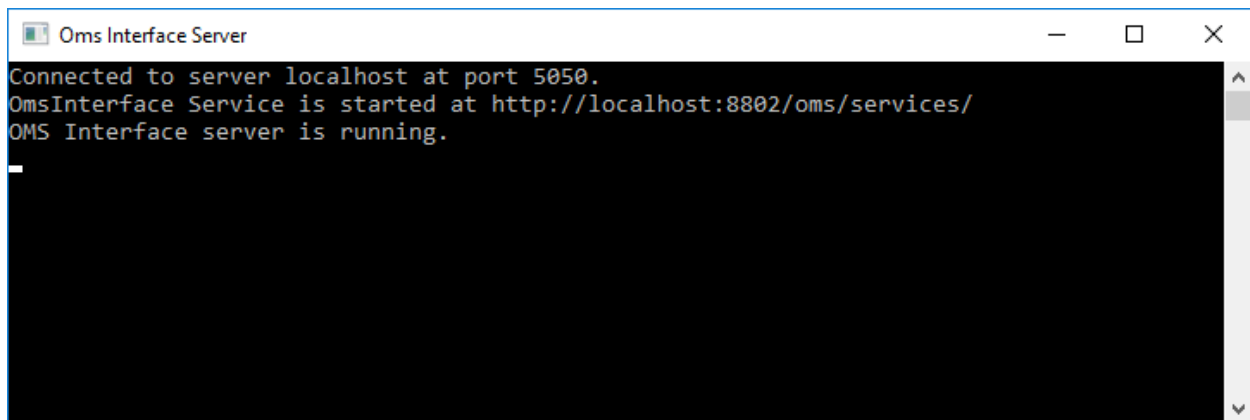


Figure 2. Running the OMS Interface Server

The OMS Interface server creates the following log files:

- **OmsInterfaceServer_<Timestamp>.txt:** Contains records about the status of the OMS Adapter process.

7.2. Running the OMS Adapter

Note

The OMS Adapter receives data from the web service, which communicates with the OMS Interface server. The OMS Interface server needs to run with administrator privileges before starting the OMS Adapter.

To start the OMS Adapter, open the Windows Start menu and navigate to All Programs> Eterra>

Distribution> ADMS Interfaces> OMS Adapter.

Note

The OMS Adapter must be run by a user with administrator privileges.

By default, the OMS Adapter starts with the OmsAdapterConfig.xml and OmsServerConfig.xml configuration files. For more details about the OMS Adapter configuration, see [ADMS Adapter Configuration Files](#). For more details about the OMS Server configuration, see the *Outage Management System Server Configuration Editor User Guide*.

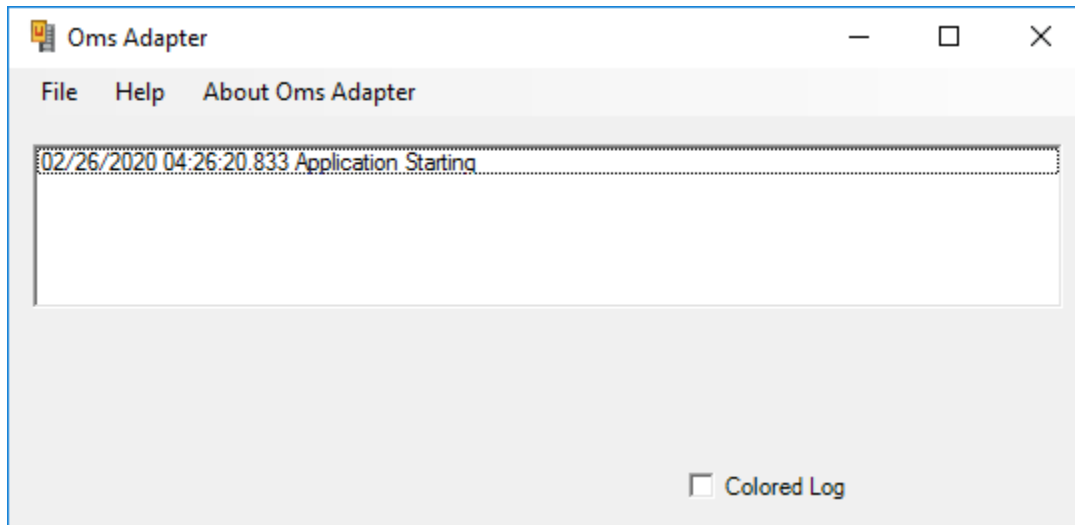


Figure 3. Running the OMS Adapter

From the File menu, you can also export Outage Assist customizations to the Outage Assist's Preferences service. For more details, refer to the *Outage Assist Software Installation and Maintenance Guide*.

The OMS Adapter creates the following log files:

- **OmsAdapter_<Timestamp>.log:** Contains records about the status of the OMS Adapter process.
- **AdapterProcessingJournal.dat:** This file is used internally to track which journal messages have been processed by the adapter engine. It does not log sending notifications to external applications.
- **SimpleFileOut.txt:** Contains notification-related data in plain text format. Used for debug purposes. Deprecated functionality.

7.3. Running the ANPS Adapter

Note

The ANPS Adapter receives data from the web service, which communicates with the OMS

Interface server. The OMS Interface server needs to run with administrator privileges before starting the ANPS Adapter.

To start the ANPS Adapter, from the Windows Start menu, navigate to All Programs> Eterra> Distribution> ADMS Interfaces> ANPS Adapter.

Note

The ANPS Adapter must be run by a user with administrator privileges.

By default, the ANPS Adapter starts with the ANPSAdapterConfig.xml configuration file. For more details about the ANPS Adapter configuration, see [ANPS Adapter Configuration](#).

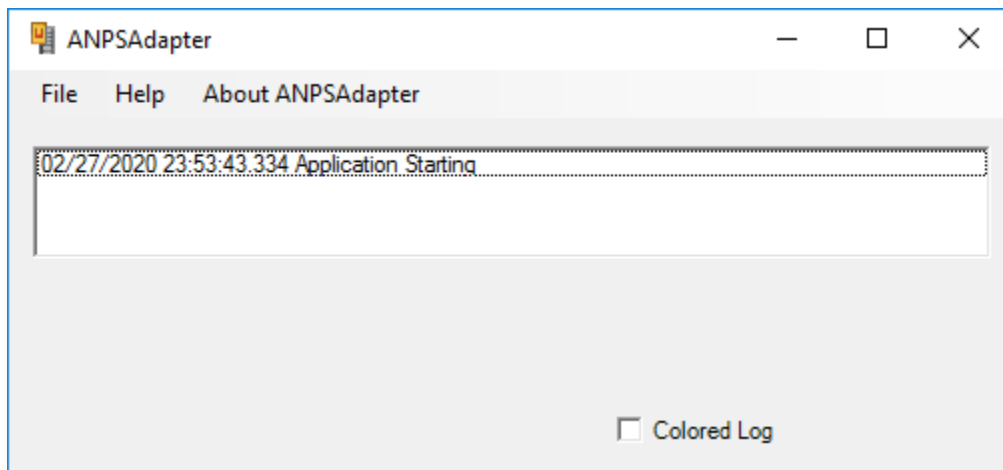


Figure 4. Running the ANPS Adapter

The ANPS Adapter creates the following log files:

- **ANPSAdapter_<Timestamp>.log:** Contains records about the status of the ANPS Adapter process.
- **ANPSAdapterProcessingJournal.dat:** This file is used internally to track which journal messages have been processed by the adapter engine. It does not log sending notifications to external applications.

7.4. Running the Customer Experience Adapter

Note

The Customer Experience Adapter receives data from the web service, which communicates with the OMS Interface server. The OMS Interface server needs to run with administrator privileges before starting the Customer Experience Adapter.

To start the Customer Experience Adapter, from the Windows Start menu, navigate to All Programs> Eterra> Distribution> ADMS Interfaces> Customer Experience Adapter.

Note

The Customer Experience Adapter must be run by a user with administrator privileges.

By default, the Customer Experience Adapter starts with the CustomerExperienceAdapterConfig.xml configuration file. For more details about the Customer Experience Adapter configuration, see [Customer Experience Adapter Configuration](#).

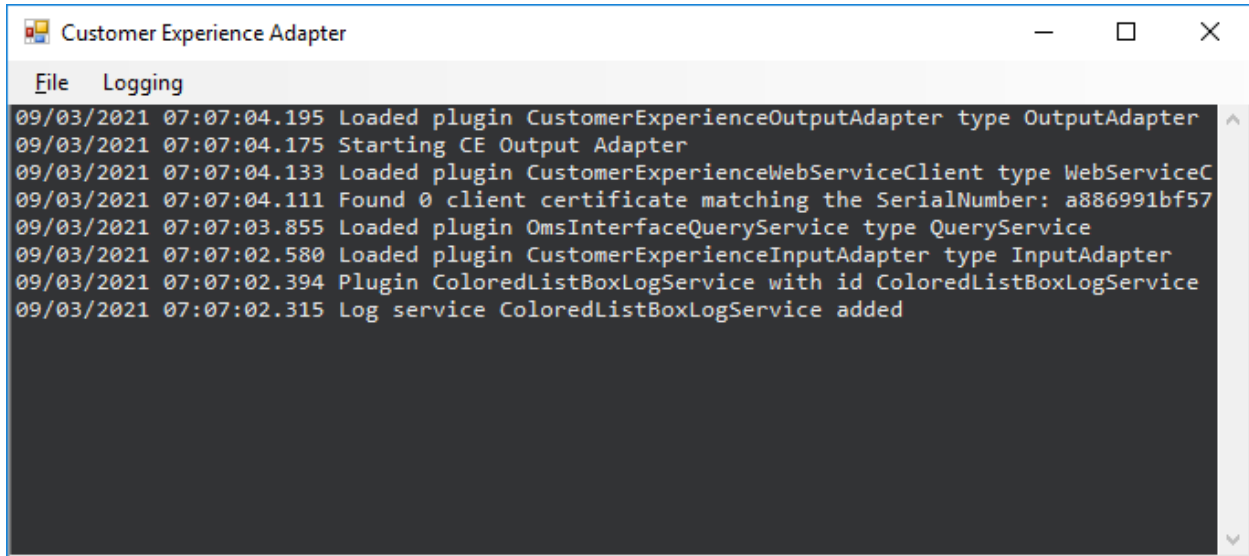


Figure 5. Running the Customer Experience Adapter

The Customer Experience Adapter creates the following log files:

- **CustomerExperienceAdapter_<Timestamp>.log:** Contains records about the status of the Customer Experience Adapter process.
- **CEAdapterProcessingJournal.dat:** This file is used internally to track which journal messages have been processed by the adapter engine. It does not log sending notifications to external applications.

7.5. Running the View Outage Portal Adapter

Note

The View Outage Portal Adapter receives data from the web service, which communicates with the OMS Interface server. The OMS Interface server needs to run with administrator privileges before starting the View Outage Portal Adapter.

To start the View Outage Portal Adapter, from the Windows Start menu, navigate to All Programs> Eterra> Distribution> ADMS Interfaces> View Outage Portal Adapter.

Note

The View Outage Portal Adapter must be run by a user with administrator privileges.

By default, the View Outage Portal Adapter starts with the ViewOutagePortalAdapterConfig.xml configuration file. For more details about the View Outage Portal Adapter configuration, see [View Outage Portal Adapter Configuration](#).

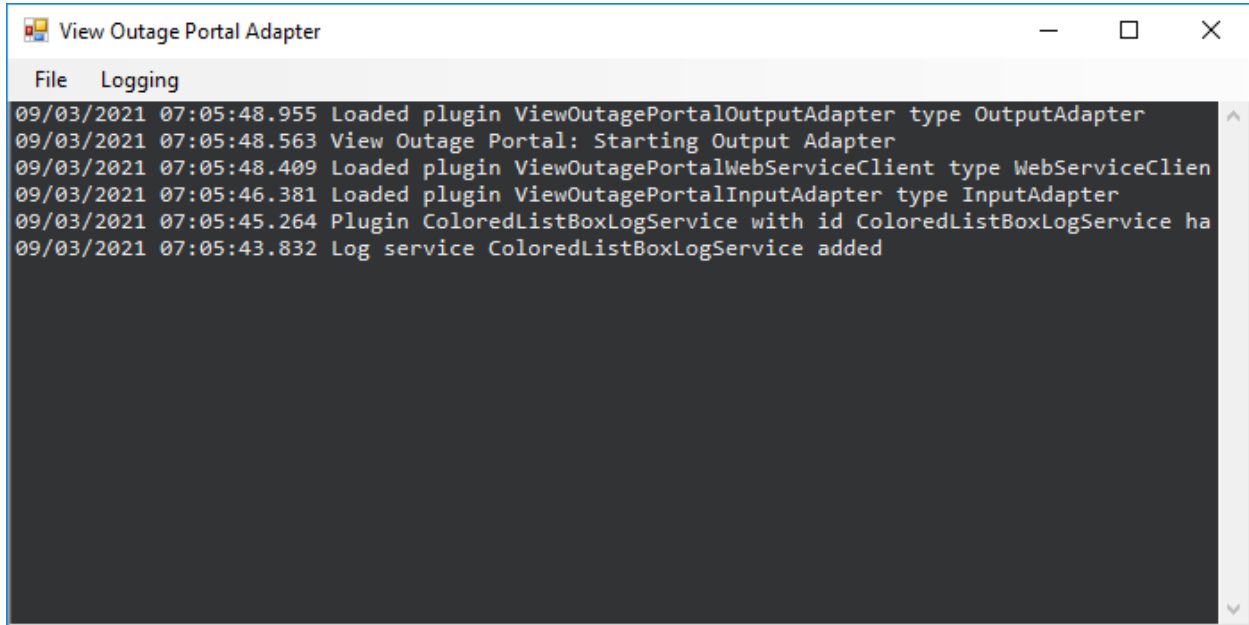


Figure 6. Running the View Outage Portal Adapter

The View Outage Portal Adapter creates the following log files:

- **ViewOutagePortalAdapter_<Timestamp>.log:** Contains records about the status of the View Outage Portal Adapter process.
- **VOPAdapterProcessingJournal.dat:** This file is used internally to track which journal messages have been processed by the adapter engine. It does not log sending notifications to external applications.

8. OMS Interface Server Configuration

The OMS Interface server reads the OmsInterfaceServerConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\client folder.

The OmsInterfaceServerConfig.xml configuration file includes the following parameters:

- **OmsServerIPs:** Name or IP address of the OMS server (NetServer) host machine. The default value is localhost.
- **OmsServerPort:** TCP/IP port number on which the listener host is listening for client connections. The default value is 5050.
- **OmsServerUseStreams:** If set to true, the OMS Server uses the streams OmsConnection interface. (This setting must be set to the same value in all configuration files.)
- **OmsServerUseTLS:** If set to true, the OMS Server uses the TLS security protocol. (This setting must be set to the same value in all configuration files.)
- **OmsServerCheckMessageSecurity:** If set to true, the OMS Server checks for security token authenticity/expiration. (This setting must be set to the same value in all configuration files.)
- **OmsServerCheckMessageSequence:** If set to true, the OMS Server checks message sequence numbers. (This setting must be set to the same value in all configuration files.)
- **OmsServerUseCredentialFile:** If set to true, the OMS Server uses a credential file for defining the private key and user SIDs. (This setting must be set to the same value in all configuration files.)
- **OmsServerCredentialInfoFile:** Path to the credential file.
- **OmsServerTimeToLive:** Time to live for OMS Server connections in seconds.
- **RequestTimeoutPeriod:** Timeout period for a request, in seconds.
- **UseBinarySerialization:** Specifies whether to use binary serialization.
- **SendCompressedMessage:** Specifies whether to send compressed messages.
- **SendBufferSize:** Buffer size for sending messages, in KB.
- **ReceiveBufferSize:** Buffer size for receiving messages, in KB.
- **EnableMessageLog:** Specifies whether to log all incoming web service requests and their responses. To use EnableMessageLog in HTTP mode, set WSSecurityEnabled to False.
- **EnableDebugLog:** Specifies whether to log detailed information for debugging purposes.
- **LogFile:** Specifies the full path to the file that is used for OMS Interface server logging purposes.
- **MaxMessagesInLogFile:** The maximum number of records in the log file. When this number is reached, another log file is created. The default value is 10,000.
- **EnableWebApi:** If set to true, the OMS Interface server web service is enabled.
- **WritePermissionValue:** The AOR write permission value. To enable the write permission checking, use the oms.Write value. To disable the write permission checking, use the blank

value.

- **WebServiceBaseURL:** Configures the base URL for trusted applications. Note that the trailing "/" is needed at the end. The default value is `http://localhost:8802/Oms/`.

Each web service application adds a specific ending to this base. For example, the URL base for the OMS Services web service is `http://localhost:8802/Oms/Services`.

Note

Use the "http" value in the URL to support both http and https modes, as appropriate. To enable secure connection, set `SecureWebService` to `True`.

- **DMSInterfaceServerURL:** Configures the endpoint to the DMS Interface Server's DMS Control service that the OMS Interface Server uses to update DMS data and request services from the DMS server. The default value is `http://localhost:8801/dms/services/`.
- **AorApiVersion:** The AOR Manager API version. Version 2 is used by default.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables retrieving a valid UAA authentication token from the UAA server.
- **UaaEndpoints:** List of endpoints for communicating with the UAA server and retrieving a valid UAA token, separated by semicolon.

Use multiple UAA token endpoints to enable the OMS Interface Server to support connection to multiple servers (Kubernetes clusters).

- **UseUAATokenForIntrospect:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token. The OMS Interface Server extracts a UAA authentication token from the authorization header of incoming requests and sends the token to the UAA server for validation.
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **UaaEndpointCheckingInSeconds:** Time period (in seconds) for checking the UAA endpoint.
- **IncidentAorSyncInSeconds:** Time period (in seconds) for requesting AOR data for all incidents from NetServer.
- **DisableServiceAuthCheck:** If this is set to true, authorization header checking for service calls is disabled. This is added for ease of testing and should not be set to true in a production system.
- **WSSecurityEnabled:** Set to false to test the OMS Interface Server calls using SoapUI or to use the `EnableMessageLog` option in HTTP mode. Make sure that `WSSecurityEnabled` values in `OmsInterfaceServerConfig.xml` and `OmsAdapterConfig.xml` are the same.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **DmsQueryUserId:** Configures the trusted user ID for the DMS Interface Server. Required for implementing DMS commands.

Note

Use the same setting for this parameter and the TRUSTED_USER_ID_DIS parameter in the NetServerMainConfig.txt configuration file.

- **TrustedUserId:** Configures the trusted user ID for the OMS Interface Server. Required for implementing OMS commands.

Note

Use the same setting for this parameter and the TRUSTED_USER_ID_OIS parameter in the NetServerMainConfig.txt configuration file.

- **CustomSecretStore:** The DMS Interface Server can retrieve the DMS Interface Server trusted user ID from the CyberArk Enterprise Password Vault so that you do not need to provide the value in the configuration file (as a plain text or encrypted string). For more information, refer to the *ADMS Software Installation and Maintenance Guide*.
 - **AssemblyName:** Name of the assembly (DLL) containing a class that implements the CyberArk secret store.
 - **ClassName:** Name of the class that implements the CyberArk secret store. This class is implemented based on the CyberArk software development kit (SDK) and supports connecting to the CyberArk API.
 - **ObjectKey:** CyberArk object key, which contains the CyberArk application ID, safe, database (folder), and object definitions of the OMS Interface Server trusted user ID defined in the Enterprise Vault Server. The format is:

<DatabaseMemberUserName>,<AccountSafe>,root,Application-<AccountPlatformID>-<AccountAddress>-<AccountUsername>
 - **UaaIdObjectKey:** CyberArk object key, which contains the CyberArk application ID, safe, database (folder), and object definitions of the OMS Interface Server UAA ID defined in the Enterprise Vault Server.
 - **UaaSecretObjectKey:** CyberArk object key, which contains the CyberArk application ID, safe, database (folder), and object definitions of the OMS Interface Server UAA secret defined in the Enterprise Vault Server.
- **OMS Interface Server Web Service Plugin Parameters:** This group includes OMS Interface Server web service plugin parameters.
 - **Enabled:** If set to true, the OMS Interface Server web service is enabled.
 - **Type:** The type that implements the OMS Interface Server web service plugin.
 - **Name:** OMS Interface Server web service plugin name.
 - **BasePath:** OMS Interface Server web service root URL.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** OMS Interface Server web service name to be shown on the user interface.

- **Version:** OMS Interface Server web service message version.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **QueryService:** The name of the OMS Interface query service.
- **PW:** Password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
- **Plugin URI Parameters:** The URI parameters in this group apply to all OMS Interface Server web service plugins.
 - **UriScheme:** URI scheme: HTTP or HTTPS.
 - **UriHost:** URI host.
 - **UriPort:** URI port.

i Note

Use different ports for different web services.

- **UriPath:** URI path.

9. ADMS Adapter Configuration Files

9.1. ANPS Adapter Configuration

The ANPS Adapter reads the ANPSAdapterConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\interfaces folder.

The ANPSAdapterConfig.xml configuration file includes the following parameters:

- **Incident Journal Folder:** Path to the folder with Incident journals.
- **ANPS Output:**
 - **ANPSEndPoint:** Configures the base URL for ANPS web services. Both HTTP and HTTPS modes are supported.
 - **UseHeaderAuthorization:** If set to True, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
 - **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
 - **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
 - **CertSubjectName:** The TLS certificate subject name's CN value.
 - **MessagePollingInterval:** Polling time for messages, in seconds.
 - **QueryService:** The name of the OMS Interface query service.
 - **BlockPostMessageTypes:** ANPS Adapter message types to block from sending to the ANPSEndPoint (for example, ANPSNotification).
 - **RejectedHttpResponseStatusCodes:** HTTP response codes that cause the associated message to be rejected and eliminated from the retry logic.
 - **RetryEnabled:** Set to true to enable retry logic.
 - **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
 - **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debug purposes.
 - **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this

functionality when using a load balancer software, as it may not support this functionality well. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every ANPS notification is sent via a new HTTP connection.
- **EnableMultiThreadPosting:** Specifies whether multi-threaded message posting is enabled.
- **PostingThreadPoolSize:** The maximum number of concurrently posting threads.
- **EnableMetrics:** Specifies whether message processing time metrics are logged.
- **PostingPublishStatisticsIntervalMin:** Time period (in minutes) for posting threads to publish statistics.
- **log4net:** Parameters for the ANPS Adapter processing log file.
 - **level:** Logging level: OFF (no logging), FATAL, ERROR, WARN, INFO, DEBUG, or ALL.
 - **appender:** Appender parameter. For more details, see <https://logging.apache.org/log4net/log4net-1.2.11/release/sdk/log4net.Appender.RollingFileAppender.html>.
- **AdapterJournalFile:** Path to the file where the ANPS Adapter internal log is recorded.
- **OMS Interface Query Service Web Service Plugin Parameters:** This group includes parameters for the OMS Interface Query Service web service plugin. This plugin supports general and predefined queries to the OMS Interface Server's Query web service.
 - **Enabled:** If set to true, the OMS Interface Query Service web service is enabled.
 - **Type:** The type that implements the OMS Interface Query Service web service plugin.
 - **Name:** The OMS Interface Query Service web service plugin name.
 - **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **OmsQueryUri:** URI for OmsQuery messages.
 - **IncidentUri:** URI for Incident messages.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
 - **LogService:** The name of the logging service.
- **OMS Interface V1 Service Web Service Client Plugin Parameters:** This group includes parameters for the OMS Interface V1 Service web service client plugin. This plugin interacts with

the OMS Interface Server's REST web service, which uses the "oms/services/rest" endpoint.

- **Enabled:** If set to true, the OMS Interface V1 Service web service client plugin is enabled.
- **Type:** The type that implements the OMS Interface V1 Service web service client plugin.
- **Name:** The OMS Interface V1 Service web service client plugin name.
- **BaseUri:** The web service root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
- **Plugin URI Parameters:** This group includes URI parameters, which apply to all ANPS Adapter web service plugins.
 - **UriScheme:** URI scheme, HTTP or HTTPS. Applies to all ANPS Adapter web service plugins.
 - **UriHost:** URI host. Applies to all ANPS Adapter web service plugins.
 - **UriPort:** URI port. Applies to all ANPS Adapter web service plugins.

Note

Use different ports for different web services.

- **UriPath:** URI path. Applies to all ANPS Adapter web service plugins.
- **FullMessages:** If set to true, then entire message payloads are written to the log file.

9.2. Customer Experience Adapter Configuration

The Customer Experience Adapter reads the CustomerExperienceAdapterConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\interfaces folder.

The CustomerExperienceAdapterConfig.xml configuration file includes the following parameters:

- **log4net:** Parameters for the Customer Experience Adapter processing log file.
 - **level:** Logging level: OFF (no logging), FATAL, ERROR, WARN, INFO, DEBUG, or ALL.
 - **appender:** Appender parameter. For more details, see <https://logging.apache.org/log4net/log4net-1.2.11/release/sdk/log4net.Appender.RollingFileAppender.html>.
- **AdapterJournalFile:** Path to the file where the Customer Experience Adapter internal log is recorded.
- **EnableFullMessageLogging:** If set to true, then entire message payloads are written to the main log file.
- **Customer Experience Adapter Plugin Parameters:** This group includes parameters for Customer Experience Adapter plugins.
 - **Enabled:** If set to true, the plugin is enabled.
 - **Type:** The type that implements the plugin.
 - **Name:** The plugin name.

- **LogService Configuration:**

LogMessageTypes:

- **Type:** Application for which the log is created.
- **SeparateLogToDisk:** If set to true, a separate log file is created for this type.
- **Color:** Set the color of log messages in the log.

LogToDisk:

- **Name:** Name of the log file.
- **EnableAppendLog:** If set to true, a log appender is added.
- **EnableDailyLog:** If set to true, a separate log file is created daily.
- **MaxEntries:** Specifies the maximum number of entries in the log file.

- **CustomerExperienceInputAdapter Configuration:**

- **DataType:** Supported data type.
- **FileFolder:** Specifies the path to the folder with incident journals.

- **CustomerExperienceWebServiceClient Configuration:**

- **BaseUri:** The CustomerExperienceWebServiceClient root URI.
- **MessageGetPollingInterval:** Polling time for CustomerExperienceWebServiceClient messages, in seconds.
- **IncidentUpdateUri:** URI for IncidentUpdateNotification messages.
- **PlannedOutageUri:** URI for PlannedOutageNotification, UpdatePlannedOutageNotification, and DeletePlannedOutageNotification messages.
- **Name:** The Customer Experience web service client name.
- **Version:** The Customer Experience web service client message version.
- **Description:** The Customer Experience web service client description.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web service runs in HTTPS mode).
- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
- **LogService:** The name of the logging service.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are

ignored for the web service plugin.

- **BlockPostMessageTypes:** CE Adapter message types to block from sending to the Customer Experience Web Service Client (for example, IncidentUpdateNotification and PlannedOutageNotification).
- **RejectedHttpResponseStatusCodes:** HTTP response codes that cause the associated message to be rejected and eliminated from the retry logic.
- **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
- **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debug purposes.
- **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using a load balancer software, as it may not support this functionality well. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every CE notification is sent via a new HTTP connection.

- **EnableMultiThreadPosting:** Specifies whether multi-threaded message posting is enabled.
- **PostingThreadPoolSize:** The maximum number of concurrently posting threads.
- **EnableMetrics:** Specifies whether message processing time metrics are logged.
- **PostingPublishStatisticsIntervalMin:** Time period (in minutes) for posting threads to publish statistics.

- **OmsInterfaceQueryService Configuration:**

- **BaseUri:** The OMS Interface Query Service root URI.
- **OmsQueryUri:** URI for OmsQuery messages.
- **IncidentUri:** URI for incident messages.
- **PlannedOutageUri:** URI for planned outage messages.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** The password value that is used to hash and place in the request authorization

header. Use the Encryption tool to encrypt the password.

- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **LogService:** The name of the logging service.
- **AcceptGZipCompression:** If set to true, the Customer Experience Adapter adds the "Accept-Encoding : gzip" entry in the HTTP request message header so that the OMS Interface Server sends gzipped content in response to the Customer Experience Adapter requests.

◦ **CustomerExperienceOutputAdapter Configuration:**

- **DataTypes:** Supported data types.
- **SendUpdateIncident:** Specifies whether to send notifications for incident updates.
- **EnableUpdateQueryBypass:** If set to true, incident and planned outage data is stored in cache and notifications use the cache data instead of the dynamically queried data. The cache is updated when incident or planned outage updates appear in the system. This also supports showing custom field updates in messages.
- **Name:** The Customer Experience Output Adapter web service name.
- **Version:** The Customer Experience Output Adapter web service message version.
- **Description:** The Customer Experience Output Adapter web service description.
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
- **BlockPredictionWaitPeriod:** The time period to wait for removing incident predictions before creating a list of customers that are no longer associated with an incident (apology customers list).
- **QueryService:** The name of the OMS Interface query service.
- **NotificationService:** The name of the Customer Experience Web Service Client.
- **LogService:** The name of the logging service.
- **EnableMomentaryOutageNotification:** Specifies whether to issue a notification about creating a momentary incident. If disabled, a notification is sent when a momentary incident becomes confirmed. If enabled, a notification is sent when a momentary incident is created.
- **EnablePostAllStatus:** If set to true, the Customer Experience Adapter sends incident update notifications for incidents in the pending clear state (for example, when changing the status to Cleared). For this feature, the application uses a cache for incident updates. This reduces queries (increases the throughput rate) but increases the number of posts. If set to false, the Customer Experience Adapter does not send incident update notifications for incidents in the pending clear state.

- **IncidentTriggers:** Specifies incident update events and field changes for which IncidentUpdateNotification notifications are triggered. Name specifies the name of the event that triggers the notification. FieldName specifies the name of the field whose update triggers the notification. NewValue optionally specifies field values that trigger the notification. To trigger notifications for a custom field change, use the CUSTOM.<CustomFieldName> field name format.
- **PlannedOutageTriggers:** Specifies planned outage update events and field changes for which PlannedOutageNotification notifications are triggered. Name specifies the name of the event that triggers the notification. FieldName specifies the name of the field whose update triggers the notification. NewValue optionally specifies field values that trigger the notification. To trigger notifications for a custom field change, use the CUSTOM.<CustomFieldName> field name format.
- **Plugin URI Parameters:** This group includes URI parameters, which apply to all Customer Experience Adapter plugins.
 - **UriScheme:** URI scheme: HTTP or HTTPS. Applies to all Customer Experience Adapter plugins.
 - **UriHost:** URI host. Applies to all Customer Experience Adapter plugins.
 - **UriPort:** URI port. Applies to all Customer Experience Adapter plugins. Use different ports for different web services.
 - **UriPath:** URI path. Applies to all Customer Experience Adapter plugins.
 - **FullMessages:** If set to true, then entire message payloads are written to the log file.

9.3. View Outage Portal Adapter Configuration

The View Outage Portal Adapter reads the ViewOutagePortalAdapterConfig.xml configuration file, which is located in the %IDMS_ROOT%\config\interfaces folder.

The ViewOutagePortalAdapterConfig.xml configuration file includes the following parameters:

- **Application Parameters**
 - **log4net:** Parameters for the View Outage Portal Adapter processing log file.
 - **level:** Logging level: OFF (no logging), FATAL, ERROR, WARN, INFO, DEBUG, or ALL.
 - **appender:** Appender parameter. For more details, see <https://logging.apache.org/log4net/log4net-1.2.11/release/sdk/log4net.Appender.RollingFileAppender.html>.
 - **AdapterJournalFile:** Path to the file where the View Outage Portal Adapter internal log is recorded.
 - **EnableFullMessageLogging:** If set to true, then entire message payloads are written to the main log file.
- **ColoredListBoxLogService Plugin Parameters:** This group includes parameters for the Colored List Box Log Service plugin.

- **Enabled:** If set to true, the plugin is enabled.
- **Type:** The type that implements the plugin.
- **Name:** The plugin name.
- ColoredListBoxLogService Configuration:

LogMessageTypes:

- **Type:** Application for which the log is created.
- **SeparateLogToDisk:** If set to true, a separate log file is created for this type.
- **Color:** Set the color of log messages in the log.

LogToDisk:

- **Name:** Name of the log file.
- **EnableAppendLog:** If set to true, a log appender is added.
- **EnableDailyLog:** If set to true, a separate log file is created daily.
- **MaxEntries:** Specifies the maximum number of entries in the log file.

- **ViewOutagePortalInputAdapter Plugin Parameters:** This group includes parameters for the View Outage Portal Input Adapter Plugin plugin.

- **Enabled:** If set to true, the plugin is enabled.
- **Type:** The type that implements the plugin.
- **Name:** The plugin name.
- **ViewOutagePortalInputAdapter Configuration:**
 - **DataType:** Supported data type.
 - **FileFolder:** Specifies the path to the folder with De-Energized Line journals.

- **ViewOutagePortalWebServiceClient Plugin Parameters:**

- **Enabled:** If set to true, the plugin is enabled.
- **Type:** The type that implements the plugin.
- **Name:** The plugin name.
- **ViewOutagePortalWebServiceClient Configuration:**
 - **BaseUri:** Configures the base URI for ViewOutagePortal web services. Both HTTP and HTTPS modes are supported.
 - **MessageGetPollingInterval:** Polling time for messages, in seconds.
 - **Name:** The View Outage Portal web service client name.
 - **Version:** The View Outage Portal web service client message version.
 - **Description:** The View Outage Portal web service client description.
 - **EnableDualSSL:** If set to True, a dual SSL web service client connection is used (the web

service runs in HTTPS mode).

- **CertStore:** The certificate store type, Personal or TrustedRoot. The certificate must be placed locally.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **MessagePollingInterval:** Polling time for messages, in seconds. This setting must match the OMS_DEENERGIZED_LINES_COLLECTION_INTERVAL_MS setting in the NetServerOmsConfig.txt file.
- **UseHeaderAuthorization:** If set to True, "authorization" header checking is enabled for incoming web service calls.
- **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
- **LogService:** The name of the logging service.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **BlockPostMessageTypes:** VOP Adapter message types to block from sending to the ViewOutagePortalEndPoint (for example, ViewOutagePortalNotification).
- **RejectedHttpResponseStatusCodes:** HTTP response codes that cause the associated message to be rejected and eliminated from the retry logic.
- **MaxPostTryCnt:** The number of times to try to deliver a message. If this count is reached for a message, the message is rejected and eliminated from the retry logic. To disable limiting the number of retries, set this parameter to "-1".
- **HttpRequestTimeoutSeconds:** Time period to wait for the client response for an HTTP request. This parameter is used for testing and debug purposes.
- **HttpDefaultRequestHeadersExpectContinue:** Specifies whether the HttpClient "ExpectContinue" behavior is enabled.

By default, DotNet enables the "ExpectContinue" behavior. This behavior results in HTTP clients posting the "Expect: 100-Continue" request header first, and then waiting for a "100-Continue" response from the server before posting the request payload. Disable this functionality when using a load balancer software, as it may not support this functionality well. Disabling this functionality eliminates one extra client server exchange for each message posted.

- **HttpKeepAlive:** Specifies whether the HttpClient "KeepAlive" behavior is enabled.

By default, DotNet enables the "KeepAlive" behavior. This behavior specifies that HTTP clients maintain the existing HTTP connection when sending messages. When this functionality is disabled, every VOP notification is sent via a new HTTP connection.

- **EnableMultiThreadPosting:** Specifies whether multi-threaded message posting is enabled.
- **PostingThreadPoolSize:** The maximum number of concurrently posting threads.

- **PostingPublishStatisticsIntervalMin:** Time period (in minutes) for posting threads to publish statistics.
- **EnableMetrics:** Specifies whether message processing time metrics are logged.
 - **ViewOutagePortalOutputAdapter Plugin Parameters:**
 - **Enabled:** If set to true, the plugin is enabled.
 - **Type:** The type that implements the plugin.
 - **Name:** The plugin name.
- **ViewOutagePortalOutputAdapter Configuration:**
 - **DataType:** Supported data type.
 - **Name:** The View Outage Portal Output Adapter web service name.
 - **Version:** The View Outage Portal Output Adapter web service message version.
 - **Description:** The View Outage Portal Output Adapter web service description.
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool to encrypt the password.
 - **NotificationService:** The name of the notification service.
 - **LogService:** The name of the logging service.
 - **SendCombinedStationSummaries:** Specifies whether the VOP Adapter sends one StationSummary object per a ViewOutagePortalNotification message or combines all StationSummary objects into one message.
 - **SendChangedStationsOnly:** If set to True, the VOP Adapter sends all current outage data from the De-Energized Line journal at start up; after that, the VOP Adapter sends incremental changes only. If set to False, the VOP adapter does not send current outage data at start up; the VOP Adapter only sends incremental changes.
- **Plugin URI Parameters:** This group includes URI parameters, which apply to all View Outage Portal Adapter plugins.
 - **UriScheme:** URI scheme: HTTP or HTTPS. Applies to all View Outage Portal Adapter plugins.
 - **UriHost:** URI host. Applies to all View Outage Portal Adapter plugins.
 - **UriPort:** URI port. Applies to all View Outage Portal Adapter plugins. Use different ports for different web services.
 - **UriPath:** URI path. Applies to all View Outage Portal Adapter plugins.
 - **FullMessages:** If set to true, then entire message payloads are written to the log file.